Depth, Oversmoothing, and Architecture: A Comparative Study of Graph Neural Networks on the Cora Citation Network

CSE 705 Deep Learning

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Abstract

Graph neural networks (GNNs) built from iterative message passing lose the ability to discriminate between nodes as depth increases, a phenomenon known as oversmoothing; this limits the receptive field available to applications such as recommendation on user-item graphs, community detection in social networks, and molecular property prediction, where relevant structure can lie several hops away. We compare four message-passing architectures: GCN, GraphSAGE, GAT, and GCNII, on the Cora citation network across depths 2, 4, 8, 16, and 32, using Dirichlet energy and mean average distance (MAD) as two complementary oversmoothing diagnostics. We capture both quantities at initialization, at the best-validation checkpoint, and at the final training epoch, which lets us separate a collapse attributable to the propagation operator from one attributable to training. Test accuracy degrades sharply beyond depth 4 for GCN, GraphSAGE, and GAT, but through distinct mechanisms rather than one shared failure: GraphSAGE and GAT do not train at all at depth 32, while GCN partially trains and develops a signature of early-layer energy collapse paired with late-layer growth. GCNII, whose identity mapping and initial-residual connection are designed against this dynamic, is the only architecture whose accuracy improves with depth. Among four tested mitigations, Jumping Knowledge restores the most deep-network performance (73.1% test accuracy at depth 32, against a 24.1% unmitigated baseline), while residual connections alone do not clearly outperform the unmitigated baseline at this depth.

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1 Introduction

1.1 Motivation

Many real-world systems are better represented as graphs than as collections of independent observations. In a citation network, for example, papers are represented as nodes and citations are represented as edges. Recommendation systems can connect users to products through ratings or purchases, while molecular graphs represent atoms and the chemical bonds between them. In each case, the relationships between objects contain useful information that would be lost if every object were considered separately.

Graph representation learning uses both node information and graph structure to learn useful numerical representations. These representations, usually called node embeddings, can then be used for tasks such as node classification, link prediction, community detection, and recommendation. Graph neural networks (GNNs) are especially useful for this purpose because they allow connected nodes to exchange information through a process known as message passing.

During message passing, each node updates its representation by combining its own features with information received from its neighbors. A single layer uses information from nearby nodes, while stacking several layers allows information to travel across a larger part of the graph. In principle, this should help a model capture relationships that are more than one step away. In practice, however, adding more layers often makes GNNs harder to train and can reduce their ability to distinguish between nodes.

Depth is useful because each additional layer increases the receptive field of a node. A deeper model can therefore collect information from nodes that are several graph steps away. This can be important in applications where relevant information is not located in the immediate neighborhood. At the same time, repeated aggregation can make node representations increasingly similar. When this happens, nodes from different classes may become difficult to separate. This behavior is commonly referred to as oversmoothing (Li et al. 2018; Oono and Suzuki 2020).

Depth is not optional in every domain that motivates this study. In physics-informed graph neural networks, which represent a discretized physical domain (a mesh, for example) as a graph and propagate a physical state across it, a useful prediction requires information to reach across enough of the domain, so oversmoothing is a first-order obstacle rather than a tunable inconvenience. Cora is small and tightly clustered compared with such a domain, and the specific numbers reported in this study are not expected to transfer to a physics mesh, which is far larger and requires information to travel much farther. What is expected to transfer is the diagnostic methodology, the measurement apparatus built around Dirichlet energy and MAD, and the qualitative behavior of the mitigations, not the numbers themselves.

1.2 The Task: Semi-Supervised Node Classification

This project studies semi-supervised node classification on the Cora citation network. Cora contains 2,708 scientific papers connected by citation links. Each paper is represented by a 1,433-dimensional binary bag-of-words feature vector and belongs to one of seven research areas (Sen et al. 2008; Yang et al. 2016). The standard public split contains 140 labeled training nodes, 500 validation nodes, and 1,000 test nodes. The features and graph positions of all nodes are available during training, but only the labels of the training nodes are used to update the model.

A two-layer Graph Convolutional Network (GCN) is known to perform well on this dataset (Kipf and Welling 2017). Reproducing that baseline is an important first step, but it does not explain what happens when the network becomes deeper. For this reason, depth is the main experimental variable in this study. We compare models with 2, 4, 8, 16, and 32 message-passing layers and examine how their classification performance and hidden representations change as more layers are added.

1.3 Thesis and Methodology

Oversmoothing is often discussed as though it were the only reason deep GNNs fail, but a decrease in accuracy does not prove that oversmoothing has occurred. A deep model may also fail because optimization becomes ineffective, gradients become unstable, or too much neighborhood information is compressed into a fixed-size representation. These mechanisms can produce similar changes in classification accuracy while affecting the hidden representations in different ways.

The purpose of this study is therefore not simply to show that deeper GNNs perform worse. Instead, we investigate how different architectures fail as their depth increases and whether the failures share the same cause. Three standard message-passing architectures are compared: GCN (Kipf and Welling 2017), GraphSAGE (Hamilton et al. 2017), and the Graph Attention Network (GAT) (Veličković et al. 2018). GCN uses degree-normalized aggregation, GraphSAGE combines a node's own representation with the mean representation of its neighbors, and GAT learns attention weights for individual connections. Because their aggregation rules differ, they may also respond differently to increasing depth.

The models are evaluated using test accuracy, micro-F1, and macro-F1. Micro-F1 is included because it is required by the project specification, although it is equal to accuracy in a single-label multiclass problem. Macro-F1 is also reported because it gives equal importance to each class and can reveal whether a model is performing poorly on less accurately predicted classes.

Classification metrics alone do not describe what is happening inside a deep network. We therefore examine the hidden node representations using two oversmoothing measures: Dirichlet energy and mean average distance (MAD). Dirichlet energy measures how much connected nodes differ from each other over the graph, while MAD measures the average cosine distance between node representations. The two metrics are used together because they capture different aspects of representation collapse.

The metrics are recorded at three stages of every run. The first measurement is taken at initialization, before any gradient update. The second is taken at the checkpoint with the lowest validation loss, which is also the checkpoint used for the reported test results. The third is taken at the final training epoch before the best checkpoint is restored. Separating these three stages helps us distinguish behavior caused by the untrained propagation process from changes that develop during training.

We also compare several methods designed to improve deep GNNs. Residual connections preserve information from earlier layers. PairNorm attempts to prevent the node representations from collapsing toward one another (Zhao and Akoglu 2020). Jumping Knowledge combines representations learned at different depths (Xu et al. 2018). GCNII uses initial residual connections and identity mapping to support much deeper graph networks (M. Chen et al. 2020). These methods are compared using both classification results and representation-level measurements.

The experiments use the same Cora split and a common training protocol across the main model comparisons. Each main configuration is repeated using ten fixed random seeds. The hidden width, activation function, dropout rate, optimizer, and model-selection rule are kept consistent wherever possible. This reduces the chance that an apparent depth effect is actually caused by a different training setup. A separate two-layer GCN experiment is also used to check that the implementation reproduces the established Cora baseline.

1.4 Findings and Scope

The proposal that initiated this project anticipated oversmoothing as a mechanism shared across architectures, with node representations expected to collapse progressively as depth increased. The results in this report support a narrower, more specific claim instead, and that narrowing is itself part of the study's contribution: representational collapse toward a low-energy state is visible cleanly at initialization, before any training occurs, but it does not survive training in the same form under this study's hyperparameters.

The central argument of this report is that depth-related failure is architecture-dependent. GCN, GraphSAGE, and GAT all lose classification performance as they become deeper, but the internal behavior of the models is not the same. The results show that deep GraphSAGE and GAT remain close to strongly collapsed initialization states and show little evidence of effective training. Deep GCN behaves differently: it trains partially and develops strong contraction in its early layers together with energy growth in later layers. GCNII does not follow either pattern and is the only tested architecture whose classification performance improves at the largest depths.

This study is limited to the Cora dataset and its standard public split. Therefore, the exact depth at which performance declines and the ranking of the mitigation methods should not be assumed to transfer directly to larger, noisier, or less homophilous graphs. The project also does not record per-layer gradient or weight norms, so some explanations of the training behavior remain supported by indirect evidence rather than measured directly. The conclusions are therefore limited to what can be established from the classification results, training curves, and representation metrics collected in the experiments.

1.5 Contributions

The main contributions of the project are:

1. A controlled comparison of GCN, GraphSAGE, and GAT across depths from 2 to 32 layers.
2. A layer-wise analysis of node representations using Dirichlet energy and MAD.
3. A comparison of representations at initialization, the selected checkpoint, and the final training epoch.
4. An empirical distinction between different depth-related failure patterns rather than treating every failure as the same form of oversmoothing.
5. An evaluation of residual connections, PairNorm, Jumping Knowledge, and GCNII as methods for improving deep GNN performance.
6. A reproducible experimental pipeline using fixed random seeds, automated result records, validation-based checkpoint selection, and generated tables and figures.

1.6 Report Roadmap

The remainder of the report is organized as follows. Section 2 reviews the background literature on graph representation learning, deep GNNs, oversmoothing, trainability, and mitigation methods. Section 3 introduces the mathematical notation, model formulations, diagnostic metrics, classification measures, and the Cora dataset. Section 4 describes the implementation and experimental design. Section 5 explains how the main design decisions are represented in the source code. Section 6 presents and discusses the experimental results. Section 7 outlines the limitations of the study and directions for future work, followed by the final conclusions.

2 Problem Statement, Review, and Background

2.1 Problem Statement

The task considered in this project is semi-supervised node classification on the Cora citation network. Each node represents a scientific paper, each edge represents a citation relationship, and the goal is to assign every paper to one of seven research categories. The difficulty is that only 140 of the 2,708 nodes are labeled for training. A successful model must therefore learn from both the words associated with each paper and the citation structure connecting the papers (Sen et al. 2008; Yang et al. 2016).

Graph neural networks are well suited to this problem because they allow information to move between connected nodes. Instead of classifying each paper from its own feature vector alone, a GNN can also use information from papers that cite it or are cited by it. This is useful on a citation network because connected papers often discuss related topics.

A shallow GNN can perform well on Cora, but this does not fully answer the question studied in this report. The main problem is to understand what happens when more message-passing layers are added. A deeper network can use information from more distant parts of the graph, but it may also become difficult to train or lose the differences between node representations. The practical question is therefore not simply whether depth helps or hurts. It is whether different GNN architectures respond to depth in the same way, and whether poor deep-model performance is actually caused by oversmoothing.

This distinction matters because several different problems can produce a similar decrease in test accuracy. A network may train successfully and then produce overly similar node embeddings. It may receive gradients that are too weak to update its earlier layers. It may compress too much distant information into a fixed-size representation, or it may stop near its randomly initialized state without learning a useful decision boundary. Looking only at the final accuracy would treat all of these situations as the same failure even though their causes and possible solutions are different.

The project addresses this problem by comparing GCN, GraphSAGE, and GAT under the same depth grid and training protocol. GCNII is also included as an architecture designed specifically for deeper graph networks. Classification metrics are examined together with layer-wise representation metrics so that a decrease in accuracy is not automatically labeled as oversmoothing.

2.2 Message Passing and the Role of Depth

Most modern GNNs can be understood through the message-passing framework (Gilmer et al. 2017). At each layer, a node collects information from its neighbors, combines that information with its current representation, and passes the result to the next layer. Although the exact aggregation rule differs between architectures, the general idea is the same.

Depth controls how far information can travel. After one message-passing layer, a node has access to information from its direct neighbors. After two layers, it can also receive information that originated from neighbors of those neighbors. Continuing this process expands the receptive field of the node.

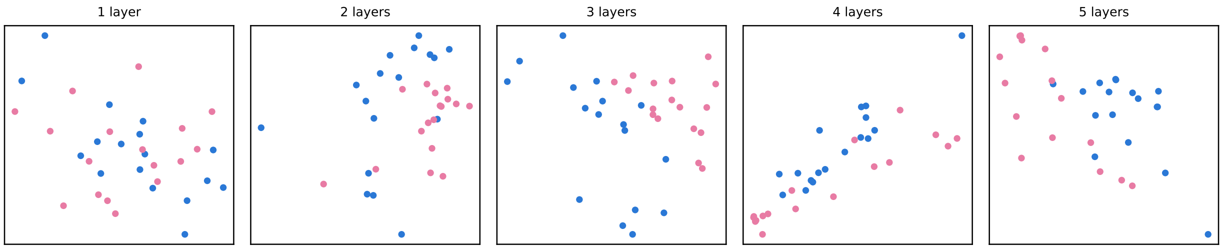
This larger receptive field is the main reason to consider deep GNNs. In many graphs, the information needed for a prediction may not be available within one or two hops. A recommendation may depend on several stages of interaction between users and items. A social-network prediction may depend on membership in a wider community rather than on immediate friendships alone. In a molecular or physical graph, distant components may also affect the property being predicted.

However, a larger receptive field is only useful when the network can preserve and organize the information it receives. Each message-passing layer mixes the representations of connected nodes. Repeating this operation many times can remove useful differences between nodes, especially when the aggregation rule behaves like an averaging process. The model then gains access to a wider neighborhood but may lose the ability to tell the nodes in that neighborhood apart.

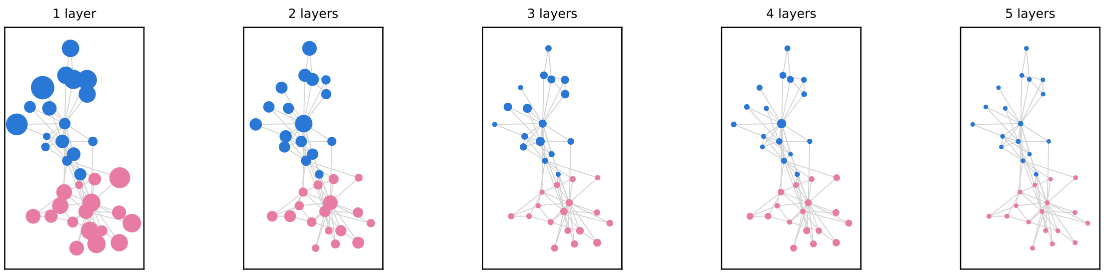
This produces the main tension studied in the project. Increasing depth can provide more structural information, but it can also weaken the class-discriminative information already present in the node embeddings.

2.3 Oversmoothing in Graph Neural Networks

Oversmoothing describes the tendency of node representations to become less distinguishable as message passing is repeated. Li, Han, and Wu connected graph convolution with Laplacian smoothing and showed that the same operation that helps nearby nodes share information can eventually make their representations too similar (Li et al. 2018). Oono and Suzuki later showed that, under suitable conditions, deep GNN representations approach a restricted subspace as depth increases, reducing the expressive power available for node classification (Oono and Suzuki 2020).



(a) Raw two-dimensional embedding-space output, no post-hoc projection, color marking the two-faction split.



(b) The same computation viewed at each node's fixed graph position (one spring layout, held constant across panels); marker size is each node's Euclidean deviation from that depth's embedding centroid, on one shared scale across all five panels so a shrinking marker is itself a collapse signal.

Figure 1: Embedding collapse in an untrained, randomly initialized GCN stack at depths 1 through 5, applied to Zachary's karate club network with one-hot node identity features. No backpropagation is performed; the illustration isolates the effect of repeated graph convolution alone, the Laplacian-smoothing mechanism (Li et al. 2018) describes.

At depth 1, the two factions are not visibly separated (Figure 1a); a single aggregation step mixes each node's identity with only its immediate neighbors', which random weights do not yet turn into a faction-aligned direction. By depth 4, the embeddings compress toward a single, nearly linear direction, a loss of rank rather than a collapse to one point, and depth 5 keeps that same low-dimensional arrangement.

This collapse is uneven across nodes, not a uniform shrinkage (Figure 1b): most nodes are already near-invisible by depth 2-3, while a handful, visibly the higher-degree ones, retain a noticeably larger marker through depth 5. Node degree correlates with retained depth-5 deviation at Pearson $r = 0.37$ (34 nodes, one seed, one graph). This is consistent with, not independent confirmation of, the $\sqrt{\text{degree}}$-weighted null-space direction the threats-to-validity discussion in Section 6 describes for GCN's own collapse on Cora.

The intuitive explanation is straightforward. Suppose two connected nodes have different feature vectors. After one aggregation step, each representation contains part of the other's information. After many similar steps, the difference between them can become much smaller. Across a complete graph, this process can cause many node embeddings to point in similar directions or to vary only within a low-dimensional space.

This becomes a classification problem when nodes from different classes are also smoothed together. The classifier receives less information with which to separate the classes, even if the graph structure was initially helpful. Oversmoothing therefore creates a depth limit: beyond a certain point, adding layers may reduce rather than improve the usefulness of the learned representations.

It is important, however, not to define oversmoothing only as a decrease in accuracy. Accuracy is an output-level measurement. It shows whether the final predictions are correct but does not show what happened inside the network. A model may have low accuracy without its hidden representations becoming similar, and a representation may become smoother without immediately causing a large accuracy decrease.

For this reason, oversmoothing should be examined directly from the hidden embeddings. Previous work has proposed measures based on pairwise distances, graph smoothness, and the geometry of the representation space. The MAD measure, for example, uses cosine distance to describe how far node representations are from one another (D. Chen et al. 2020). Dirichlet energy instead measures variation over the graph and becomes smaller when connected nodes move toward a smooth representation. The mathematical definitions of both measures are given in Section 3.

The two metrics do not measure exactly the same property. MAD is mainly concerned with the directions of node embeddings, while Dirichlet energy also depends on how the representations vary relative to the graph. A set of embeddings can become nearly one-dimensional without reaching the null space of the graph Laplacian. In that case, MAD may indicate strong collapse while Dirichlet energy remains nonzero. Using both measures provides a more careful view than relying on either one alone.

2.4 Oversmoothing, Trainability, and Oversquashing

Oversmoothing is only one possible difficulty in a deep GNN. Another is trainability. Adding layers creates a longer path between the loss and the earlier model parameters. If useful gradient information does not reach those parameters, the early layers may remain close to their initial values. The model may then produce poor predictions because it did not train effectively, not because training gradually pushed its embeddings into an oversmoothed state.

Kipf and Welling report a depth study in an appendix rather than in their main results: testing depths from one to ten layers, with and without residual connections, they find the best performance at two to three layers and training becoming difficult past roughly seven layers without residual connections on citation networks (Kipf and Welling 2017). This is a statement about optimization difficulty, not about representation collapse, and it is the specific prior-art finding this study's central architecture-dependent result builds on. The appendix experiment uses five-fold cross-validation over all labeled nodes rather than the 140-label standard split used throughout this study, so its numbers are qualitative precedent for a trainability limit at depth, not a directly comparable baseline.

This difference can be difficult to identify from one final checkpoint. A network that barely moved from initialization and a network that trained before collapsing may both have low accuracy and low embedding diversity. Their solutions would be different. The first may require changes to optimization, initialization, or gradient flow. The second may require a mechanism that preserves representation differences during propagation.

Oversquashing is another related but separate problem. As depth increases, the number of nodes that can influence a representation may grow rapidly. All of this information must still be compressed into a hidden vector of fixed width. Important signals from distant nodes may therefore be weakened or lost before they reach the target node (Topping et al. 2022). Oversquashing concerns the compression and transmission of distant information, while oversmoothing concerns the loss of distinction among node representations.

These mechanisms can occur together. A deep model may have similar node embeddings, weak gradient flow, and difficulty transmitting distant information at the same time. The terms should nevertheless not be treated as interchangeable. Doing so would make it impossible to determine which intervention is appropriate.

The experimental design in this project helps separate these possibilities by capturing the representations at three stages. The initialization capture shows the behavior of the untrained network. The selected-checkpoint capture shows the state used to produce the reported test results. The final-epoch capture shows where training ended before the best checkpoint was restored. Comparing these stages gives evidence about whether a failure was already present at initialization or developed during optimization.

2.5 Architectural Differences

The project compares architectures that use different forms of message passing. These differences provide a useful test of whether depth-related failure is a general property of GNNs or whether it depends on the aggregation rule.

GCN uses a symmetrically normalized adjacency operator to combine the representations of neighboring nodes (Kipf and Welling 2017). The normalization accounts for node degree and makes the operation efficient on a sparse graph. At the same time, repeated normalized aggregation has a clear smoothing effect, which makes GCN an important model for studying oversmoothing.

GraphSAGE was introduced as an inductive framework that learns an aggregation function rather than a separate embedding for every node (Hamilton et al. 2017). Its mean-aggregation form treats the central node and the average of its neighbors through separate transformations. GraphSAGE can also use sampled neighborhoods, which makes it more practical than full-batch GCN on large graphs. In this project, Cora is small enough for full-graph training, so GraphSAGE is mainly used to study how its aggregation rule behaves with depth.

GAT replaces fixed degree-based aggregation weights with learned attention coefficients (Veličković et al. 2018). In principle, this allows the model to give more weight to useful neighbors and less weight to irrelevant ones. However, attention does not guarantee that deep representations remain distinct. A GAT still combines information repeatedly, and the attention mechanism introduces additional parameters and optimization complexity.

These architectures are not expected to have identical depth behavior. Their normalization rules, parameterizations, and initializations differ. A fair comparison must therefore hold the surrounding experimental conditions as constant as possible while recognizing that a shared training setup may not be the individually optimal setup for every architecture.

2.6 Approaches for Improving Deep GNNs

Several methods have been proposed to make deep GNNs easier to train or less prone to representation collapse. They work through different mechanisms, so their effects should not be judged from accuracy alone.

Residual connections allow a layer to retain information from an earlier representation rather than replacing it completely. This creates a direct path through the network and can improve gradient flow. It may also reduce the loss of node-specific information during repeated aggregation. A residual connection does not directly control the distance between node embeddings, however, so it does not guarantee that oversmoothing will be prevented.

PairNorm takes a more direct approach by normalizing the node embeddings so that their overall dispersion does not continuously shrink (Zhao and Akoglu 2020). Instead of changing the graph convolution itself, it changes the geometry of the representation after each layer. This makes it particularly relevant when the main problem is that embeddings are moving too close together.

Jumping Knowledge avoids relying only on the deepest representation (Xu et al. 2018). It combines information from several layers, allowing the final prediction to use both shallow and deep features. Shallow layers retain more local information, while deeper layers contain information from larger neighborhoods. A model can therefore use the depth that is most useful without discarding every earlier representation.

GCNII changes the architecture more fundamentally. It repeatedly reintroduces the initial node representation and uses an identity-mapping term to limit how strongly each layer transforms its input (M. Chen et al. 2020). These choices are designed to preserve the original features and support much deeper graph networks. Unlike residual connections, PairNorm, and Jumping Knowledge in this project, GCNII is treated as a separate architecture rather than as a hook attached to a standard GCN.

The methods also introduce different costs. Jumping Knowledge requires a readout that combines several layers. GCNII uses additional projections and architecture-specific parameters. PairNorm adds normalization but no learned weights. Residual connections are simple when the layer widths match but may require projections otherwise. These differences mean that classification accuracy, representation behavior, model capacity, and computational cost all need to be considered when the methods are compared.

2.7 Large-Scale and Noisy Graphs

Cora is useful for controlled experiments because it is small enough to train in full batch and to measure every node representation at every layer. Many real graphs are much larger. A recommendation network, transaction graph, or social platform may contain millions of nodes and edges. In these settings, storing the full graph and all intermediate embeddings in memory may not be possible.

Neighbor sampling is one common response to this problem. GraphSAGE, for example, can train on sampled neighborhoods instead of expanding every neighbor at every layer (Hamilton et al. 2017). This reduces memory and computation, but it also changes the message-passing process. A node may receive a different sampled neighborhood from one training step to another. Metrics such as Dirichlet energy and MAD must then be interpreted carefully because the graph used for training may not be the same graph used for measurement.

Large graphs also make the depth problem more important. In a small, well-connected citation network, several hops may already cover a substantial part of the graph. In a large sparse graph, a useful signal may genuinely be many hops away. A shallow architecture may not reach it, while a deep architecture may oversmooth, oversquash, or become too difficult to optimize before the signal arrives.

Noise creates another challenge. Real citation, social, recommendation, and transaction networks may contain missing, incorrect, or irrelevant edges. Message passing does not automatically know whether an edge is reliable. Once incorrect information is aggregated, additional layers can spread it farther through the graph. Noisy node features can create a similar effect because their influence is shared with neighboring nodes.

GAT may appear naturally suited to noisy graphs because it can assign different weights to neighbors, but learned attention is not a guarantee that unreliable edges will be ignored. Attention coefficients are learned from the same limited data as the rest of the model and may themselves be unstable. Normalization, robust aggregation, graph denoising, and uncertainty-aware edge weighting are possible directions, but they are outside the empirical scope of the present Cora study.

The exact numerical findings from Cora should therefore not be assumed to transfer directly to large or noisy graphs. What may transfer more reliably is the evaluation method: compare initialized and trained representations, use more than one smoothing metric, keep the representation band comparable across architectures, and distinguish failure to optimize from collapse that develops during training.

2.8 Gap Addressed by This Study

The same accuracy decline with depth can represent different internal failures, and a method that helps one failure may not help another; accuracy alone does not distinguish them.

This study addresses that gap through a controlled comparison with three main features.

First, architecture and depth are examined together: GCN, GraphSAGE, and GAT are evaluated across the same depths. GCNII provides an additional comparison with an architecture designed specifically for depth.

Second, the study combines output-level and representation-level evidence. Accuracy and macro-F1 describe predictive performance; Dirichlet energy and MAD describe changes in the hidden embeddings.

Third, the representation metrics are captured before training, at the selected checkpoint, and at the final epoch, which makes it possible to ask whether a deep model learned and then deteriorated or whether it remained close to a poor initial state.

The scope remains deliberately limited. The project studies one dataset, one public split, and a coordinated training protocol. It does not claim to identify a universal depth limit for GNNs or to establish that one mitigation is always best. Its goal is narrower: to show how a careful combination of architecture comparison, training evidence, and representation diagnostics can give a more accurate account of why deep GNNs fail.

3 Theory and Datasets

This section introduces the mathematical ideas used in the project and explains how they relate to the Cora node-classification task. The goal is to define the graph, the learning problem, the four GNN architectures, and the measurements used to study what happens as the networks become deeper.

3.1 Graph Notation

A graph is written as

$$G = (V, E),$$

where V is the set of nodes and E is the set of edges. In the Cora citation network, each node is a scientific paper and each edge represents a citation relationship.

Let

$$N = |V|$$

be the number of nodes. The graph structure can be stored in an adjacency matrix

$$A \in \{0, 1\}^{N \times N},$$

where

$$A_{ij} = \begin{cases} 1, & \text{if nodes } i \text{ and } j \text{ are connected,} \\ 0, & \text{otherwise.} \end{cases}$$

Although citations are directed in the original data, the processed Cora graph used in this project is treated as undirected for message passing. Therefore, if node i is connected to node j, information can travel in both directions.

Each node also has an input feature vector. The full node-feature matrix is

$$X \in \mathbb{R}^{N \times d},$$

where d is the number of features and the ith row, x_i, contains the features of node i. For Cora, these features form a binary bag-of-words representation of each paper.

The class label of node i is written as

$$y_i \in \{1, \dots, C\},$$

where C is the number of classes. Cora contains seven research-topic classes.

The nodes are divided into three disjoint sets:

$$V_{\text{train}}, \qquad V_{\text{val}}, \qquad V_{\text{test}}.$$

The training labels are used to update the model. The validation labels are used for model selection and early stopping. The test labels are kept separate and are used only to report the final performance.

3.2 Self-Loops and Graph Normalization

A GNN should usually preserve a node's own information when it gathers information from its neighbors. For this reason, self-loops are added to the adjacency matrix:

$$\widetilde{A} = A + I,$$

where I is the identity matrix.

The degree of node i in the augmented graph is

$$\tilde{d}_i = \sum_{j=1}^{N} \widetilde{A}_{ij}.$$

The corresponding degree matrix is

$$\widetilde{D} = \operatorname{diag}\left(\tilde{d}_1, \dots, \tilde{d}_N\right).$$

A common normalized propagation matrix is

$$\widehat{A} = \widetilde{D}^{-1/2} \widetilde{A} \widetilde{D}^{-1/2}.$$

This normalization reduces the effect of differences in node degree. Without it, nodes with many neighbors could produce messages with much larger magnitudes than nodes with only a few neighbors.

The normalized graph Laplacian is

$$\mathcal{L} = I - \widehat{A}.$$

The Laplacian measures how much a signal changes across the graph. A signal that has similar values on connected nodes has low Laplacian energy, while a signal that changes strongly across edges has high energy.

3.3 The Null Space of the Normalized Laplacian

The null space of the normalized Laplacian is important for understanding oversmoothing. For a connected graph,

$$\mathcal{L} \widetilde{D}^{1/2} \mathbf{1} = 0,$$

where $\mathbf{1}$ is the all-ones vector. This means that the zero-energy direction is proportional to

$$\widetilde{D}^{1/2} \mathbf{1}.$$

The entries of this vector are the square roots of the augmented node degrees. Therefore, on an irregular graph, a completely smooth representation under the normalized Laplacian is not necessarily constant across nodes. Instead, it is weighted by the square root of node degree.

This distinction matters in this project because Cora is not a regular graph. Its nodes do not all have the same degree. A representation may become one-dimensional and point in the constant direction while still having nonzero normalized Dirichlet energy. This is one reason why Dirichlet energy and MAD are both needed: the two metrics can respond differently to the same representation geometry.

A direct consequence follows for how "collapse" should be read under this metric. A representation that fully collapses under Dirichlet energy, in the sense of converging onto this null space rather than merely becoming similar across nodes, is converging toward a representation proportional to the square root of node degree, not toward a single constant vector. High-degree nodes therefore retain a larger-magnitude representation than low-degree nodes even at total collapse under this metric. "Collapsed," in this report, does not mean that every node's representation becomes numerically identical.

3.4 Semi-Supervised Node Classification

A GNN with parameters θ receives the feature matrix and graph structure and produces one output vector for each node:

$$Z = f_\theta(X, A),$$

where

$$Z \in \mathbb{R}^{N \times C}.$$

The row Z_i contains the class scores, or logits, for node i. The logits are converted into class probabilities using the softmax function:

$$\widehat{Y}_{ic} = \frac{\exp(Z_{ic})}{\sum_{k=1}^{C} \exp(Z_{ik})}.$$

The predicted class is

$$\hat{y}_i = \underset{c \in \{1,\ldots,C\}}{\arg\max}\, \widehat{Y}_{ic}.$$

Although predictions are produced for every node, the training loss is calculated only on the labeled training nodes. Using one-hot encoded labels Y_{ic}, the cross-entropy loss is

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{|V_{\mathrm{train}}|} \sum_{i \in V_{\mathrm{train}}} \sum_{c=1}^{C} Y_{ic} \log\left(\widehat{Y}_{ic}\right).$$

This setting is called semi-supervised because only a small portion of the nodes have labels available for training.

It is also transductive. The features and graph positions of the validation and test nodes are available while the model is trained, but their labels do not contribute to the loss.

3.5 Message Passing

Most GNNs can be described using a message-passing framework (Gilmer et al. 2017). Let $h_i^{(\ell)}$ be the representation of node i at layer ℓ. The input representation is

$$h_i^{(0)} = x_i.$$

A general message-passing layer first collects information from the neighbors of node i:

$$m_i^{(\ell)} = \mathrm{AGGREGATE}^{(\ell)}\left(\left\{h_j^{(\ell)} : j \in \mathcal{N}(i)\right\}\right).$$

The node representation is then updated:

$$h_i^{(\ell+1)} = \mathrm{UPDATE}^{(\ell)}\left(h_i^{(\ell)}, m_i^{(\ell)}\right).$$

The aggregation function should not depend on the order in which the neighbors are listed. Sum, mean, maximum, and attention-weighted sums are all examples of permutation-invariant aggregation methods.

After one layer, a node can use information from its direct neighbors. After two layers, it can use information from nodes that are two graph steps away. In general, an L-layer model can receive information from an approximately L-hop neighborhood.

This growing receptive field is the main benefit of depth. However, each layer also mixes connected representations. Repeating this process can reduce the differences among nodes and lead to oversmoothing.

3.6 Graph Convolutional Network

The Graph Convolutional Network uses the normalized propagation matrix introduced above (Kipf and Welling 2017). A GCN layer is written as

$$H^{(\ell+1)} = \sigma\left(\widehat{A} H^{(\ell)} W^{(\ell)}\right),$$

where

- $H^{(\ell)}$ contains the node representations at layer ℓ;
- $W^{(\ell)}$ is a trainable weight matrix;
- $\widehat{A}$ performs normalized neighborhood aggregation; and
- σ is an activation function such as ReLU.

The initial matrix is

$$H^{(0)} = X.$$

The multiplication by $\widehat{A}$ mixes each node's representation with the representations of its neighbors. The multiplication by $W^{(\ell)}$ then learns how the aggregated features should be transformed.

GCN is simple and efficient, but repeated multiplication by $\widehat{A}$ has a smoothing effect. As the number of layers increases, the representations can move toward the low-energy space of the normalized Laplacian (Li et al. 2018; Oono and Suzuki 2020).

3.7 GraphSAGE

GraphSAGE separates the representation of the central node from the representation collected from its neighbors (Hamilton et al. 2017). For mean aggregation,

$$m_i^{(\ell)} = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} h_j^{(\ell)}.$$

The update can be written as

$$h_i^{(\ell+1)} = \sigma\left(W_{\text{self}}^{(\ell)} h_i^{(\ell)} + W_{\text{neigh}}^{(\ell)} m_i^{(\ell)} + b^{(\ell)}\right).$$

Here, the node's own representation and the average neighbor representation have separate trainable transformations.

This differs from GCN, where the self and neighbor contributions are mixed through one normalized propagation operator. GraphSAGE was originally introduced for inductive learning and supports neighbor sampling, although full-graph training is used for Cora in this project. Figure 2 shows the sampling, aggregation, and combination steps in sequence.

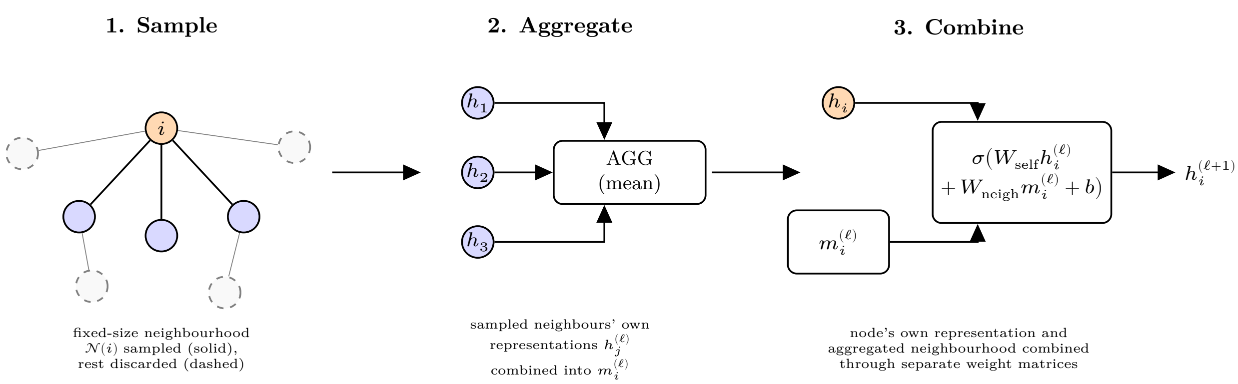


Figure 2: The GraphSAGE sample-and-aggregate pipeline: a fixed-size neighborhood is sampled around each node, the sampled neighbors' own representations are aggregated (mean, here), and the result is combined with the node's own representation through separate weight matrices, recreated as original artwork from the concept in Hamilton, Ying, and Leskovec (2017), Figure 1.

3.8 Graph Attention Network

The Graph Attention Network learns how much importance to assign to each neighbor (Veličković et al. 2018). First, the node representation is linearly transformed:

$$q_i^{(\ell)} = W^{(\ell)} h_i^{(\ell)}.$$

For node i and one of its neighbors j, an unnormalized attention score is

$$e_{ij}^{(\ell)} = \text{LeakyReLU}\left(\left(a^{(\ell)}\right)^T \left[q_i^{(\ell)} \| q_j^{(\ell)}\right]\right),$$

where $\|$ represents concatenation.

The scores are normalized across the neighborhood using softmax:

$$\alpha_{ij}^{(\ell)} = \frac{\exp\left(e_{ij}^{(\ell)}\right)}{\sum_{k \in \mathcal{N}(i) \cup \{i\}} \exp\left(e_{ik}^{(\ell)}\right)}.$$

The new representation is

$$h_i^{(\ell+1)} = \sigma\left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \alpha_{ij}^{(\ell)} q_j^{(\ell)}\right).$$

A larger value of $\alpha_{ij}^{(\ell)}$ gives node j more influence on the new representation of node i.

GAT can use several attention heads. Each head learns its own attention coefficients and transformed features. The head outputs are usually concatenated in hidden layers and may be averaged in the output layer.

Attention makes the aggregation more flexible, but it does not remove the depth problem. A deep GAT still performs repeated neighborhood aggregation and can still produce similar node representations. Figure 3 shows the coefficient computation for a single neighbor alongside the multi-head aggregation.

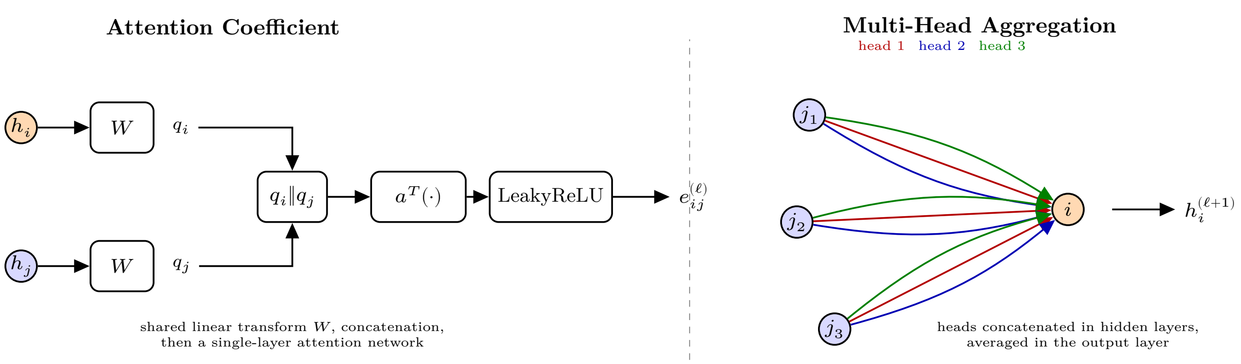


Figure 3: Left, the attention-coefficient computation between a node and one neighbor: a shared linear transform, concatenation, and a single-layer attention network followed by LeakyReLU. Right, multi-head attention aggregation: each head computes its own coefficients, and the head outputs are concatenated in hidden layers or averaged in the output layer. Recreated as original artwork from the concept in Veličković et al. (2018), Figure 1.

3.9 GCNII

GCNII was designed to support deeper graph convolutional networks (M. Chen et al. 2020). It introduces two main ideas: an initial residual connection and an identity mapping.

A simplified GCNII layer is

$$H^{(\ell+1)} = \sigma\left(\left[(1-\alpha_\ell)\widehat{A}H^{(\ell)} + \alpha_\ell H^{(0)}\right]\left[(1-\beta_\ell)I + \beta_\ell W^{(\ell)}\right]\right).$$

The first bracket mixes the propagated hidden representation with the initial representation $H^{(0)}$. This allows the original node information to remain available even in deep layers.

The second bracket combines an identity mapping with a trainable transformation. This limits how strongly each layer can change its input and also provides a more direct path for gradient flow.

GCNII is treated as a separate architecture in this project because its update rule is different from a standard GCN with an added normalization or readout method.

3.10 Mitigation Mechanisms

Three additional methods are applied to the standard architectures in the mitigation experiments.

3.10.1 Residual Connections

A residual connection adds the previous representation to the output of the current layer:

$$H^{(\ell+1)} = F^{(\ell)}\left(H^{(\ell)}\right) + H^{(\ell)}.$$

The direct path helps preserve information from earlier layers and can improve gradient flow. An identity residual can only be applied when the two matrices have the same shape. Figure 4 shows the two routes the representation takes across one layer.

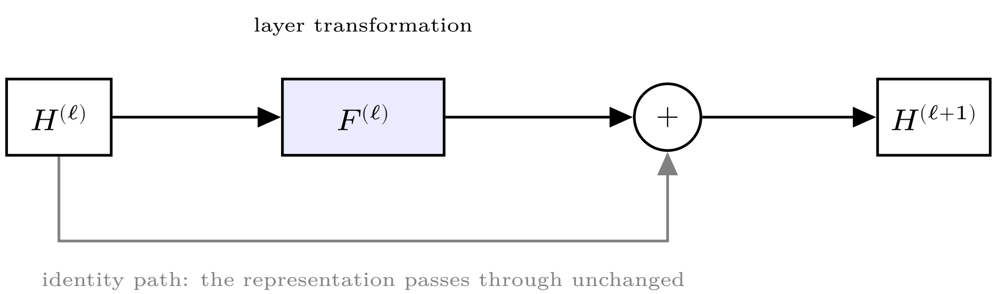


Figure 4: A residual connection around one layer. The representation $H^{(\ell)}$ reaches the addition node by two routes: the layer transformation $F^{(\ell)}$, and an identity path that carries it through unchanged. Their sum forms $H^{(\ell+1)}$, so the earlier representation stays available to the next layer even when the layer transformation contributes little.

3.10.2 PairNorm

PairNorm directly controls the spread of node representations (Zhao and Akoglu 2020). The version used in this project first centers the embeddings:

$$\widetilde{H} = H - \frac{1}{N}\mathbf{1}\mathbf{1}^T H.$$

Each node representation is then normalized separately:

$$h_i^{\mathrm{PN}} = s\frac{\widetilde{h}_i}{\|\widetilde{h}_i\|_2 + \varepsilon},$$

where s is a scale parameter and ε prevents division by zero. The authors treat s as data-dependent and select it per dataset in their reference implementation, which offers $\{0.1, 1, 10, 50\}$ as candidate values (Zhao 2019).

This is the scale-individual version of PairNorm (PN-SI). It prevents the rows of the embedding matrix from continuously shrinking toward one another. Figure 5 shows the two steps applied to a small set of row vectors.

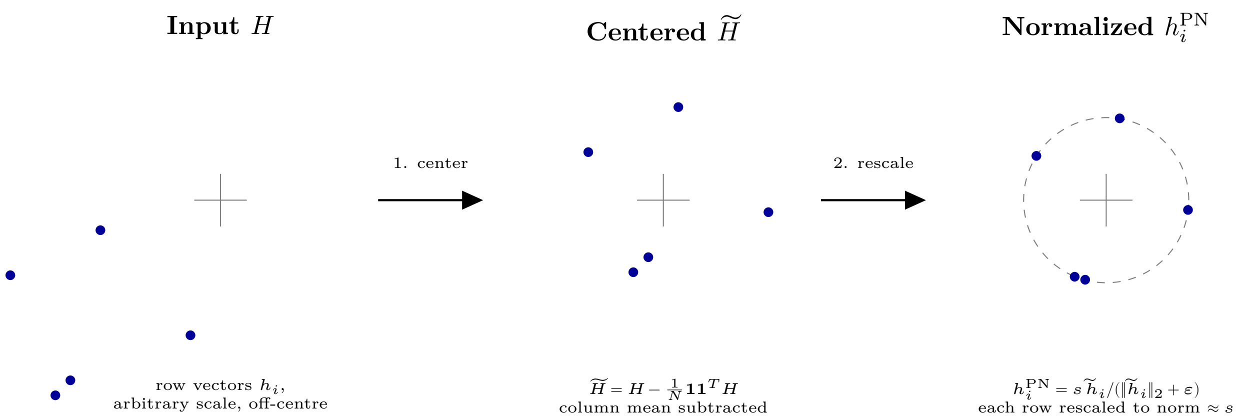


Figure 5: The two-step PairNorm procedure applied to five illustrative row vectors: centering by subtracting the column mean, then per-row rescaling to a fixed norm s. Recreated as original artwork from the concept in Zhao and Akoglu (2020), Figure 2.

3.10.3 Jumping Knowledge

Jumping Knowledge uses representations from several network depths rather than relying only on the final layer (Xu et al. 2018). In this project, the learned hidden representations are combined using element-wise maximum pooling:

$$H_{\mathrm{JK}} = \max\left(H^{(1)}, H^{(2)}, \dots, H^{(L)}\right),$$

where the maximum is taken separately for each node and hidden feature.

The pooled representation is then passed through a linear output layer. This allows the classifier to use information that appeared in an earlier layer even if later layers have become less useful. Figure 6 shows the per-layer paths into the shared aggregation.

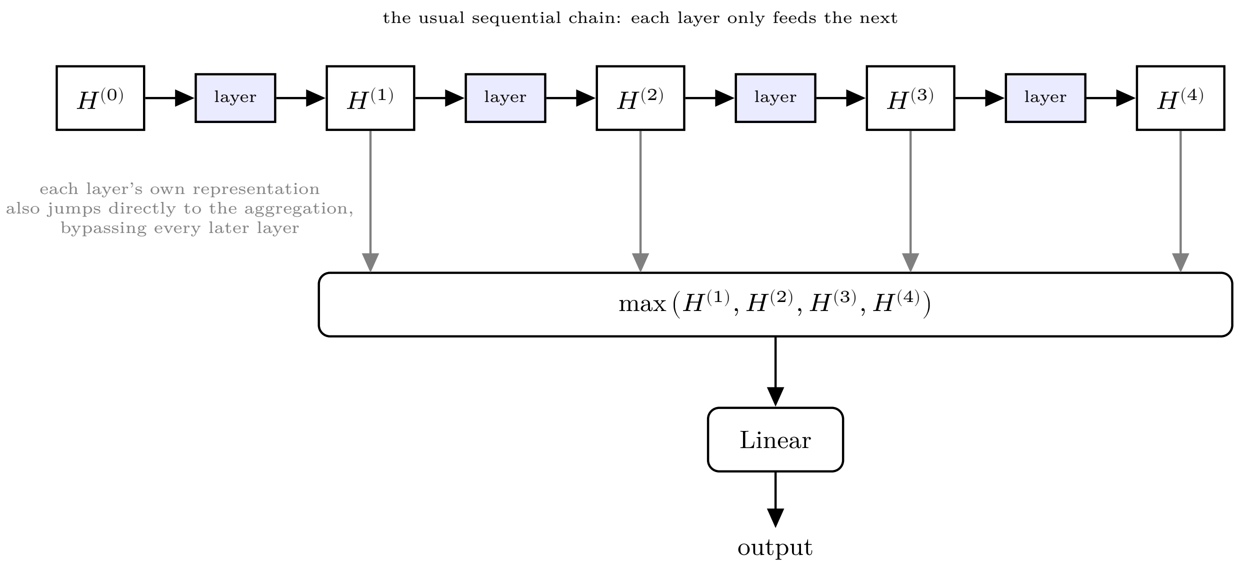


Figure 6: Per-layer representations feeding into a shared aggregation (max-pooling, here) before the final linear readout. Each layer's own representation jumps directly to the aggregation rather than passing only through the next layer, illustrating why the method is called "jumping" knowledge.

3.11 Hidden Representations

For a model with L message-passing layers, the stored representation sequence is

$$H^{(0)}, H^{(1)}, \ldots, H^{(L)}.$$

The matrix $H^{(0)}$ is the original input feature matrix. The later matrices are the learned layer representations.

In this project, a hidden representation is measured after the graph convolution, mitigation hook, and activation function, but before dropout. This represents the deterministic hidden representation passed toward the next layer without including the random dropout mask.

In the implementation these matrices are the list `layerEmbeddings`, indexed the same way, so the prose term and the code identifier name one object.

The purpose of measuring these matrices is to examine how their geometry changes across depth. Oversmoothing is expected to reduce the differences among their rows, but different metrics capture different forms of this change.

3.12 Three Metric Capture Points

The hidden representations are recorded at three stages of each run.

3.12.1 Initialization

The first capture is taken before any gradient update. It describes the behavior of the randomly initialized network.

This capture helps answer whether a deep architecture already contracts its representations before training begins.

3.12.2 Selected Checkpoint

The second capture is taken after the weights with the lowest validation loss have been restored. These are the same weights used to report test accuracy and macro-F1.

This is the most important capture for relating representation behavior to classification performance.

3.12.3 Final Epoch

The third capture is taken using the weights from the final training epoch, before the selected checkpoint is restored.

This capture shows where the optimizer ended. Comparing it with the selected checkpoint reveals whether the model continued to improve, remained unchanged, or deteriorated after its best validation state.

The three captures help separate two situations that could otherwise look similar. A network may start in a collapsed state and fail to train, or it may train and later develop a different representation pattern. A single checkpoint cannot clearly distinguish between these cases.

3.13 Dirichlet Energy

Dirichlet energy measures how much a representation changes over the graph. In this project, the graph used for measurement includes self-loops:

$$\widetilde{G} = (V, \widetilde{E}),$$

with augmented adjacency matrix $\widetilde{A}$ and augmented degrees $\tilde{d}_i$.

For an embedding matrix H, the normalized-Laplacian Dirichlet energy is

$$E(H) = \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \widetilde{A}_{ij} \left\| \frac{h_i}{\sqrt{\tilde{d}_i}} - \frac{h_j}{\sqrt{\tilde{d}_j}} \right\|_2^2 .$$

The same quantity can be written in matrix form as

$$E(H) = \mathrm{Tr}\left(H^T \mathcal{L} H\right).$$

Self-loop terms have zero direct contribution because a node is identical to itself. However, self-loops still change the augmented degrees and therefore change the normalization applied to every edge.

Layers in this project can have different widths. To avoid giving a higher-dimensional layer a larger energy simply because it has more features, the energy is divided by the embedding width:

$$E_{\dim}(H) = \frac{E(H)}{\dim(H)}.$$

A small energy means that the representation changes little across graph edges. A decreasing energy curve is consistent with graph smoothing. However, energy alone does not prove that every node representation is identical, because the zero-energy direction depends on node degree.

3.14 Mean Average Distance

MAD measures the directional separation between node representations (D. Chen et al. 2020). The cosine distance between nodes i and j is

$$D_{ij} = 1 - \frac{h_i^T h_j}{\|h_i\|_2 \|h_j\|_2 + \varepsilon}.$$

A cosine distance close to zero means that the two representations point in nearly the same direction. A larger value means that they are more separated.

For each node, the mean of its nonzero cosine distances is calculated. Let

$$\mathcal{P}_i = \left\{ j : j \neq i \text{ and } D_{ij} > 0 \right\}.$$

The row-level mean is

$$\mathrm{MAD}_i = \frac{1}{|\mathcal{P}_i|} \sum_{j \in \mathcal{P}_i} D_{ij}.$$

The final MAD is the mean over rows that contain at least one nonzero distance:

$$\mathrm{MAD}(H) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \mathrm{MAD}_i,$$

where

$$\mathcal{I} = \{ i : |\mathcal{P}_i| > 0 \}.$$

The diagonal is excluded because the distance from a node to itself should not affect the result. If every pairwise distance is zero, MAD is defined as zero rather than left undefined.

MAD and Dirichlet energy are complementary. MAD measures whether embeddings point in similar directions, while Dirichlet energy measures whether the embeddings are smooth with respect to the fixed graph and its degree normalization.

Chen et al. also define MADGap, the difference between a node's mean distance to remote, multi-hop neighbors and its mean distance to nearby ones, and show that it correlates with test accuracy closely enough to serve as a training-free predictor of model quality (D. Chen et al. 2020). This study does not compute MADGap; only MAD is reported. A metric validated as an accuracy predictor risks blurring the distinction this study is built to keep separate, between a representation having collapsed and a model performing badly, since MADGap's own justification is that the two track each other. Reporting MAD alone keeps the representation-level measurement and the classification result independent, so that either can be read without presupposing the other.

3.15 Comparable Hidden-Layer Band

Not every stored representation should be compared in the same layer-wise curve.

The input matrix $H^{(0)}$ is excluded because it contains sparse 1,433-dimensional bag-of-words features rather than learned hidden representations. A direct comparison between this input and a 64-dimensional hidden layer would mix changes in representation type, width, and scale with the effect of message passing.

For models using the standard last-layer readout, the final representation is also excluded. It contains seven-dimensional class logits rather than hidden features. Cross-entropy training directly shapes this decision space, so its energy measures a mixture of classification behavior and graph smoothing.

The comparable band is therefore formed from the contiguous learned hidden layers that share the model's main hidden width.

For a standard last-layer readout, the band is

$$\mathcal{B} = \{1, 2, \dots, L-1\}.$$

For Jumping Knowledge, the final convolution also produces a hidden-width representation before the separate readout. Its comparable band is therefore

$$\mathcal{B}_{\mathrm{JK}} = \{1, 2, \dots, L\}.$$

Defining this band from the actual tensor widths prevents different architectures from being compared using representation spaces that do not have the same meaning.

3.16 Relative Energy Curve

Raw energy values can differ because of weight initialization, architecture, and representation magnitude. To focus on the change across layers, energy is reported relative to the first layer in the comparable band.

Let

$$\ell_0 = \min(\mathcal{B}).$$

The relative energy at layer ℓ is

$$R_\ell = \frac{E_{\mathrm{dim}}\left(H^{(\ell)}\right)}{E_{\mathrm{dim}}\left(H^{(\ell_0)}\right)}.$$

Therefore,

$$R_{\ell_0} = 1.$$

A value below one means that the energy has decreased relative to the first comparable hidden layer. A value above one means that the energy has increased.

This ratio is useful for comparing the shapes of energy curves across models, but it should not be interpreted as a complete measure of embedding collapse. The MAD curve and the original energy values are also needed.

3.17 Contraction Slope

A single summary value is obtained by fitting a line to log-energy across the comparable band. For each $\ell \in \mathcal{B}$,

$$\log\left(\max\left\{E_{\dim}\left(H^{(\ell)}\right), \varepsilon_E\right\}\right) \approx a + b\ell,$$

where

$$\varepsilon_E = 10^{-12}.$$

The coefficient b is called the contraction slope.

- A negative slope means that energy generally decreases with depth.
- A slope near zero means that energy remains relatively stable.
- A positive slope means that energy generally grows across the measured layers.

The energy floor prevents the logarithm of zero. Its consequence is that very strong contraction or growth can appear less extreme after values reach the floor. The floor can pull the estimated slope toward zero, but it cannot create a trend that was not already present.

At least two comparable layers are required to fit a slope. A two-layer model with a standard last-layer readout has only one layer in its comparable band. Its contraction slope is therefore undefined rather than equal to zero.

The slope summarizes a curve, but it does not replace the curve. Two models can have similar fitted slopes while showing different behavior in their early and late layers.

3.18 Connection to Oversmoothing Theory

A common theoretical view of oversmoothing describes repeated graph propagation as a contraction toward the low-frequency space of the graph Laplacian. Under simplifying assumptions, the layer energies can satisfy a bound of the form

$$E\left(H^{(\ell+1)}\right) \leq (1-\lambda)^2 \left\|W^{(\ell)}\right\|_2^2 E\left(H^{(\ell)}\right),$$

where λ represents a graph spectral quantity and $\|W^{(\ell)}\|_2$ is the spectral norm of the layer weight matrix (Cai and Wang 2020).

The bound explains two important points.

First, the graph operator can contract the representation from one layer to the next. Repeating this contraction can produce an approximately exponential decrease in energy.

Second, the layer weights also affect the result. Large weight norms may reduce, cancel, or reverse the contraction caused by graph propagation. For this reason, energy growth in a trained model is not automatically inconsistent with oversmoothing theory. The trainable transformations are part of the mechanism, not a separate source of noise.

The experiments do not test the numerical tightness of this bound because the spectral gap and layer weight norms are not recorded across the full sweep. The bound is used to motivate the log-energy analysis and the interpretation of the contraction slope.

3.19 Classification Metrics

3.19.1 Accuracy

Test accuracy is the proportion of test nodes classified correctly:

$$\text{Accuracy} = \frac{1}{|V_{\text{test}}|} \sum_{i \in V_{\text{test}}} \mathbf{1}\left(\hat{y}_i = y_i\right).$$

Accuracy gives equal weight to each test node.

3.19.2 Precision and Recall

For class c, precision is

$$P_c = \frac{TP_c}{TP_c + FP_c},$$

and recall is

$$R_c = \frac{TP_c}{TP_c + FN_c}.$$

Here, TP_c, FP_c, and FN_c are the true positives, false positives, and false negatives for class c.

The class-specific F1 score is

$$F_{1,c} = \frac{2P_cR_c}{P_c + R_c}.$$

3.19.3 Micro-F1

Micro-F1 first combines the counts from all classes:

$$F_{1,\text{micro}} = \frac{2P_{\text{micro}}R_{\text{micro}}}{P_{\text{micro}} + R_{\text{micro}}}.$$

For single-label multiclass classification, micro-F1 is mathematically equal to accuracy. Every incorrect prediction creates one false positive for the predicted class and one false negative for the correct class. Micro-F1 is still reported because it is part of the project requirements.

3.19.4 Macro-F1

Macro-F1 averages the class-specific F1 scores:

$$F_{1,\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} F_{1,c}.$$

Each class contributes equally, regardless of how many test examples it contains. Macro-F1 is therefore useful for identifying models that perform reasonably in overall accuracy but concentrate most of their predictions into only a small number of classes.

3.19.5 Reference Baselines

Two reference values give the accuracy and macro-F1 numbers reported in Section 6 a floor to compare against. The uniform-random rate is the accuracy a classifier achieves by guessing among the C classes with equal probability and no information about the input:

$$\text{Accuracy}_{\text{random}} = \frac{1}{C}.$$

For Cora's seven classes, this is approximately 14.3%.

The majority-class rate is the accuracy achieved by a fixed classifier that always predicts whichever class is most frequent in the test split, without using any node feature or graph information. Unlike the uniform-random rate, this floor depends on the actual class distribution of the test split rather than on the number of classes alone; on the standard split used throughout this study, the majority class accounts for 319 of the 1,000 test nodes, a majority-class floor of 31.9%. A trained model whose test accuracy falls below this value is performing worse than a rule that ignores the input entirely, which is a stronger and more specific claim than a low accuracy number on its own.

3.20 The Cora Citation Network

Cora is a standard benchmark for semi-supervised node classification (Sen et al. 2008; Yang et al. 2016). It contains

- 2,708 scientific papers;
- 1,433 binary input features per paper; and
- seven research-topic classes.

The seven classes are:

1. Case-Based Reasoning;
2. Genetic Algorithms;
3. Neural Networks;
4. Probabilistic Methods;
5. Reinforcement Learning;

6. Rule Learning; and
7. Theory.

Each feature indicates whether a vocabulary term appears in a paper. The original feature vectors are sparse because each paper contains only a small part of the full vocabulary.

The graph connects papers through citation relationships. The processed Planetoid version used in this project treats these connections as undirected for message passing.

The public split contains:

Split	Number of nodes	Purpose
Training	140	Used to calculate the loss and update the model
Validation	500	Used for checkpoint selection and early stopping
Testing	1,000	Used only for final evaluation

The 140 training nodes contain 20 labeled papers from each of the seven classes. The remaining nodes still participate in message passing even though their labels are not used in the training loss.

The graph's degree distribution is highly skewed: node degree in the augmented graph ranges from 2 to 169, with a mean of approximately 4.9. This skew is not a decorative statistic. It is the reason the augmented normalized Laplacian's null space diverges from the constant vector on Cora, discussed above: on a regular graph, where every node has the same degree, the square-root-of-degree direction and the constant direction coincide, but Cora's wide degree range keeps them apart, and Section 6 depends on this distinction directly.

Cora also exhibits homophily: papers tend to cite other papers on related topics, so a node's label is correlated with its neighbors' labels more often than chance alone would predict. This is what makes neighborhood aggregation an informative operation for this task in the first place; on a graph without homophily, combining a node's representation with its neighbors' would not, on average, help classify it.

3.21 Feature Normalization

The feature vectors are row-normalized before training. For node i,

$$\tilde{x}_i = \frac{x_i}{\sum_{k=1}^{d} x_{ik} + \varepsilon}.$$

After normalization, the nonzero features in each row sum to approximately one. This prevents papers containing more vocabulary terms from automatically having feature vectors with much larger total magnitudes.

The same normalization is used for every architecture and depth. It is therefore a fixed part of the data pipeline rather than an experimental variable.

3.22 Graph Used by the Models and Metrics

The dataset loader returns the undirected graph without self-loops. The GNN layers add their required self-loops and normalization internally.

The oversmoothing metrics build a separate augmented copy of the graph:

$$\widetilde{A} = A + I.$$

The same augmented metric graph is used for GCN, GraphSAGE, GAT, and GCNII. This is important for a cross-architecture comparison. If each architecture were measured using a different graph operator, a difference in energy could come from either the embeddings or the measuring graph.

For GAT, this means that Dirichlet energy is measured relative to the fixed Cora graph rather than the model's learned attention-weighted graph. The result describes how GAT's embeddings vary over Cora's original structure. It does not describe variation over a different attention graph at every layer.

3.23 Summary

The project treats a GNN layer as both a feature transformation and a graph propagation step. Increasing the number of layers expands the receptive field, but it may also change the node representations in ways that reduce their usefulness for classification.

GCN, GraphSAGE, GAT, and GCNII use different update rules, so they are not expected to respond to depth in exactly the same way. Residual connections, PairNorm, and Jumping Knowledge also address different parts of the problem.

Classification accuracy and macro-F1 show whether the predictions remain useful. Dirichlet energy and MAD show how the hidden representation changes. The three capture points separate initialization behavior from behavior that develops during training. Finally, the comparable band and contraction slope allow the layer-wise energy curves to be compared without mixing raw inputs, hidden representations, and output logits.

4 Implementation Details

This section describes what was built and why, at the level of design decisions rather than line by line; Section 5 describes how the code expresses those decisions.

4.1 Setup

`data.LoadCora` loads the public-split Cora citation network through PyTorch Geometric's `Planetoid` dataset (Fey and Lenssen 2019) and row-normalizes the node features (Section 3.21), matching Kipf and Welling's published GCN setup (2017, sec. 5.2). An invariant check confirms the loaded graph's shapes and raises, rather than warns, if any fails, so a PyG version upgrade that silently altered the graph's shape is caught immediately. Every run is seeded on the integer list 0 through 9, with the same seed controlling both the model's random initialization and the training stochasticity that follows, so a reported mean and standard deviation over ten seeds reflects ten independent draws rather than ten runs sharing one initialization. Every run also records its own environment at write time; all 534 records carry Python 3.14.4, PyTorch 2.13.0 (CPU build), PyTorch Geometric 2.8.0, and device `cpu`.

4.2 Model Implementation

Section 3 gives four update rules. The implementation realizes all four as one parameterized model, in which `convType` selects the convolution and everything else (depth, width, mitigations, and the point at which each layer's representation is recorded) is identical across architectures.

Mitigations attach to the shared layer loop by composition rather than inheritance, so the unmitigated baseline and every mitigated variant execute the identical control-flow path and a measured difference between them is attributable to the mitigation itself (Section 5.3's `models` entry gives the full reasoning).

Hidden width is 64 for every architecture, rather than each architecture's own published width (16 for GCN; 8 heads of 8 for GAT). Capacity parity is the precondition for comparing architectures at matched depth: at width 16, GAT's eight heads would each attend over two-dimensional keys, close to degenerate, placing a confound in the study's headline figure that has nothing to do with depth. The fidelity arm (Section 6.2) recovers the published width as a separately measured quantity.

4.2.1 Keeping Depth Comparable Across Architectures

Two numerical facts distinguish GCNII's per-layer computation from the other three architectures':

Architecture	Uses $H^{(0)}$ at every layer	Extra layers outside counted depth L
GCN	No	None
GraphSAGE	No	None
GAT	No	None
GCNII	Yes	2 (input projection, output projection)

Depth is defined as a message-passing hop count, and GCNII is made to match that definition exactly. Its two required linear projections sit outside the counted depth L, so GCNII's L layers are L real message-passing hops. Depth is the study's primary comparison axis, and this is what makes GCNII's L the same quantity as the others'. The known cost: those two projections give GCNII two extra parameterized layers the other three architectures do not have, bounded in size and not scaling with depth, which can only overstate GCNII's capacity relative to its peers, at every depth (revisited in Section 6.7).

4.2.2 Recording the Hidden Representation

Where the hidden representation is recorded within a layer follows Section 3.11; Figure 7 places that position inside the layer. Three figures in Section 6 are computed from the recorded tensors: Figure 10, Figure 11, and Figure 14.

- **Before dropout**, because inverted dropout rescales surviving units by $1/(1-p)$ at train time, which would inflate raw Dirichlet energy by a constant that the energy ratio absorbs but that MAD does not respond to identically, making the two metrics disagree for a reason that is an artifact rather than a finding.

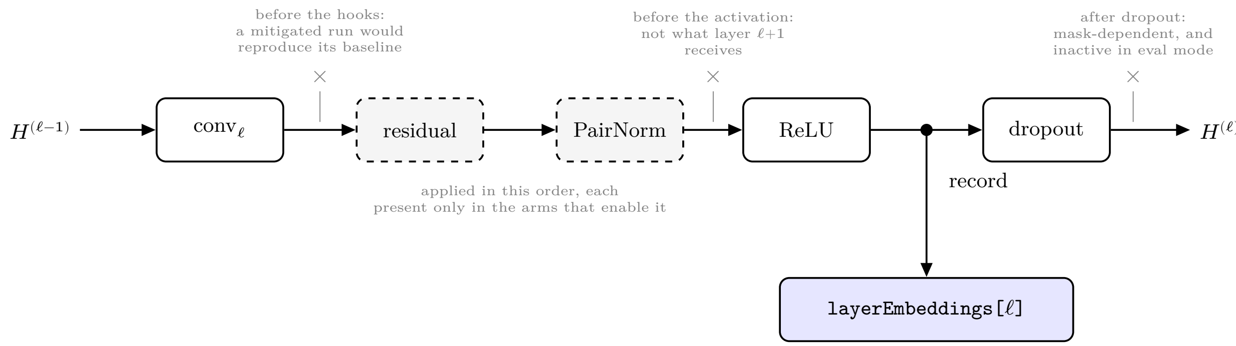


Figure 7: One layer's interior and the point at which its representation is recorded. The recording point sits after the convolution, after every mitigation hook, and after the activation, but before dropout. The three crossed positions are the alternatives this subsection rejects, each labeled with the reason. Hook boxes are dashed because a hook is present only in the arms that enable it; their order is fixed at residual before PairNorm regardless of the order in which the mitigation names are supplied.

4.3 Mitigation Implementations

The three mechanisms specified in Section 3.10 (residual connections, PairNorm, and Jumping Knowledge) attach to the base model independently of which architecture is running. Our proposal listed GCNII as a fourth mitigation; we implement it instead as a fourth convolution type, since it modifies the layer's update rule rather than wrapping an existing one.

Mechanism	What it does to the representation	Learned parameters	Implementation choice
Residual	Adds the previous layer's output back into the current layer's output, so each layer starts from what came before instead of replacing it	None	Plain identity addition on interior layers; skipped where input and output widths differ
PairNorm	After each layer, recenters all node representations and rescales each to a fixed length, holding their total spread constant as depth grows	None	PN-SI variant, scale fixed at 1.0, untuned
Jumping Knowledge	Keeps every layer's representation and combines them at the readout, so the final prediction is not forced to rely on the deepest layer alone	One linear readout	Element-wise max pooling, one of the three aggregations in Xu et al. (2018)

Hook ordering is fixed at residual before PairNorm, regardless of the order the mitigation names are supplied in. Residual is part of the update rule and defines what the layer's output is; PairNorm then controls the spread of that output (Section 3.10), which requires it to act last.

4.3.1 Residual

Residual is implemented as identity addition, following the depth study in Kipf and Welling's Appendix B (Kipf and Welling 2017). The mechanism of Section 3.10 adds a layer's input to its output, which requires both to carry the same number of features per node. Layer 1 maps 1433 input features onto the hidden width of 64, and under the standard last-layer readout the last layer maps 64 onto the seven classes, so the addition is skipped at both boundaries and fires on interior layers alone.

4.3.2 PairNorm

PairNorm is applied after each convolution, and in this study only to GCN. We use PN-SI (Section 3.10), which the authors' usage guidance pairs with GCN: their repository reports that plain PairNorm behaves badly under the symmetric normalized adjacency GCN uses (Zhao 2019). A later note in the same repository traces part of that behavior to a bug and suggests plain PairNorm may now suffice, while stating this has not been fully tested, so we retain PN-SI. We hold s at 1.0 rather than selecting it per dataset, because the coordinated search (Section 4.6) holds one configuration fixed across every architecture and depth; tuning PairNorm's scale alone would compare a tuned mitigation against untuned baselines. Whatever PairNorm scores at this setting is therefore a floor: tuning s could raise it, not lower it.

4.3.3 Jumping Knowledge

Jumping Knowledge aggregates the stored hidden representations by element-wise max pooling and projects the result to the output width with a single linear layer. Max pooling holds the readout's input width at the hidden width regardless of depth; concatenation, one of the three aggregations in Xu et al. (2018), would make it $T \times \text{hiddenDim}$ for T pooled layers, so the JK arm's parameter count would vary along the study's comparison axis (Section 4.2.1). JK's comparable band, $1..L$ rather than $1..L-1$, follows directly from the band-derivation rule in Section 3.15.

4.4 Training and Evaluation Harness

4.4.1 Optimizer and Weight Decay

Every run uses PyTorch's Adam implementation (`torch.optim.Adam`) (Paszke et al. 2019) at the configuration the search in Section 4.6 selected (`learningRate=0.01`, `dropout=0.5`, `weightDecay=5e-4`), the same values Kipf and Welling publish. Weight decay is applied uniformly across every layer's parameters, which keeps the regularization from becoming an implicit function of depth (Section 4.2.1).

4.4.2 Early Stopping and Checkpoint Selection

Early stopping and checkpoint selection are both driven by validation loss. Training halts when validation loss has not improved for `patience` consecutive epochs; the reported and captured checkpoint is the epoch of minimum validation loss (Section 3.12.2), whose weights are restored after the loop ends. Kipf and Welling specify a stopping rule (200 epochs maximum, halting if validation loss does not decrease within a window of 10) (Kipf and Welling 2017), and this study adds the selection rule, tying it to the same signal so `bestEpoch` is unambiguous. Validation loss and validation accuracy are recorded at every epoch in `trainingCurve`.

4.4.3 Patience

Patience is 100 epochs, held constant across every architecture, depth, mitigation arm, and seed, with `maxEpochs=1000` as a ceiling. A 32-layer network may sit on a plateau for many epochs before its loss begins to descend, or may never descend. At Kipf and Welling's window of 10, a run whose validation loss stops improving after the first epoch halts near epoch 12 (Section 6.5), leaving a flat training curve that a non-converging optimization and a genuinely untrainable architecture would share. Holding patience constant across depth keeps the depth comparison valid (Section 4.2.1).

4.4.4 Three Metric Captures

Three metric captures are taken per run, each an explicit eval-mode forward pass under `torch.no_grad()` (Section 3.12): epoch 0, before any gradient step; the final epoch, on the weights the loop ended with; and the selected checkpoint, after the best `state_dict` is restored. The order matters, and Figure 8 places the three on the sequence of one run. The final-epoch capture is taken while the loop's own weights are still loaded, before the checkpoint weights are restored. Taken afterward, both captures would read the same weights, and `finalMetrics` would duplicate `checkpointMetrics` on every run. What the two record separately is how the model changed over the epochs between its best validation loss and the end of the loop (Section 6.5).

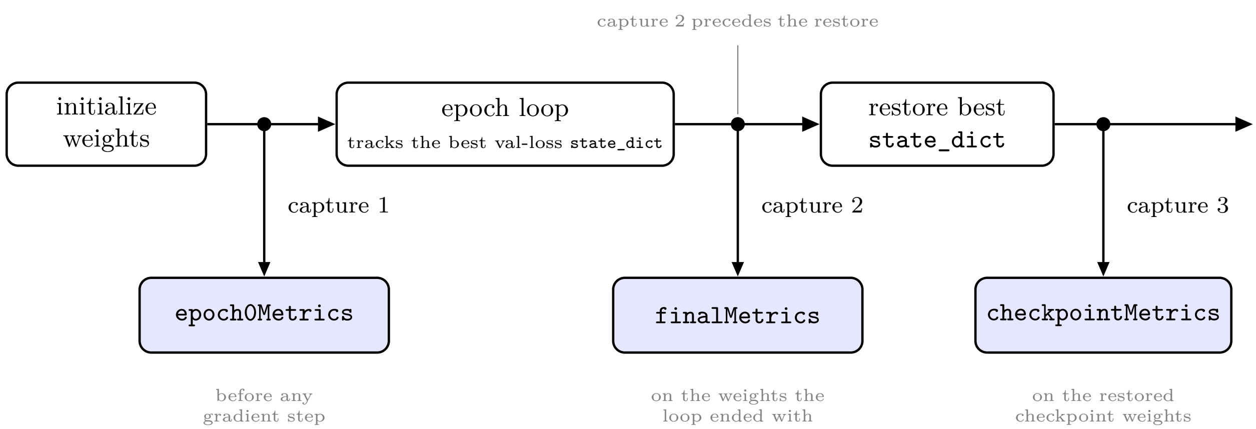


Figure 8: The order of the three metric captures within one run. Each is a separate eval-mode forward pass. The capture on the final epoch's weights is taken before the best-validation `state_dict` is restored.

4.5 Metrics Implementation

Each choice below holds the measurement instrument fixed against something the study varies: architecture, split, layer width, or depth.

4.5.1 The Fixed Metric Graph

Dirichlet energy and MAD are computed on the same graph for all three architectures: the augmented graph $\widetilde{G}$ of Section 3.22. The three aggregate over their neighbors differently, GCN by a fixed symmetric normalization and GAT by attention weights that are learned and change at every layer and every epoch, and the metric graph does not follow those differences.

The metric graph is a measuring instrument rather than part of the model under test, and holding it fixed makes the embeddings the only thing that varies across the cross-architecture comparison. Matching it to GAT's own propagation graph is ill-defined in the first place, since GAT's attention re-weights the graph per layer and throughout training. Energy at consecutive layers would then be measured against different graphs, making the fitted contraction slope compare quantities with different units. A model could also lower measured energy purely by concentrating attention with representations otherwise unchanged, so the metric would partly measure itself. For GAT, the fitted decay rate describes GAT's embeddings with respect to Cora's fixed graph structure, not with respect to GAT's own attention-weighted propagation, legitimate and necessary for a cross-architecture comparison, but not the exact quantity Cai and Wang's bound (2020) describes for an attention-based architecture.

4.5.2 Dirichlet Energy and MAD

Both metrics are computed over all 2,708 nodes; neither consumes labels, so the train, validation, and test masks of Section 3.20 play no part.

Dirichlet energy is per-dimension, $E(H)$ divided by the layer's width, which removes the effect of comparing a 1433-dimensional input against a 64-dimensional hidden representation, and is computed as Section 3.13 defines it. Self-loop terms contribute zero to the energy sum but change every node's augmented degree, which is why the energy is computed on the augmented graph (Section 3.2).

MAD follows the reduction convention of Eq. 1-4 in D. Chen et al. (2020) (Section 3.14): a cosine-distance matrix, averaged over non-zero entries only, at both the per-row stage and the final stage. Two further details:

- The diagonal is excluded before reduction. For a row with positive norm this changes nothing: a row's self-cosine-similarity is 1, so it already contributes a zero distance that the non-zero filter drops. It matters for an all-zero row, where the row-norm clamp (Section 5.3) makes the self-similarity compute as 0 rather than 1, leaving a spurious non-zero self-distance that would survive the filter.
- Both reduction denominators return 0 rather than `NaN` when a count is 0, the case a fully collapsed configuration produces.

4.5.3 Contraction Slope and Relative Energy

The contraction slope of Section 3.17 is computed once per capture, by least squares on `log(energy)` against layer index over the comparable band of Section 3.15. Energies are floored at $\varepsilon_E = 10^{-12}$ before the fit, since collapsed layers reach zero energy in float32 and the logarithm of zero is undefined.

For the unmitigated depth-32 GCN baseline (`gcn_none_d32_s0`) the floor binds on 6 of 31 band layers at epoch 0, where the reported slope is `-0.7579` against an unfloored `-0.8508`, and on 22 of 31 at the checkpoint, where it is `+0.8622` against `+1.1435`. The floor pulls a reported slope toward zero whichever direction it runs, so both reported values understate the magnitude of change.

Where the band has fewer than two points the fit returns `nan`, following Section 3.17. At depth 2 under the standard last-layer readout the band is a single index, so every depth-2 configuration outside the Jumping Knowledge arm carries no slope; JK's band of $1..L$ admits a fit at that depth.

The relative energy curve of Section 3.16 divides each band layer's energy by the energy at $\ell_0 = \min(\mathcal{B})$, the first layer in the band. Section 6 reports its endpoint, `E_last/E_1`.

Values that come out `NaN` or infinite, a zero reference energy for instance, are plotted untouched rather than dropped, so a collapsed run shows a visible gap in a curve rather than a fabricated point.

The energy curve is divided by layer width only; the squared Frobenius norm $\|H_l\|_F^2$ is recorded per layer, so a magnitude-normalized variant remains derivable at aggregation time. Cai and Wang's bound, $E_{l+1} \leq (1-\lambda)^2\|W\|^2 E_l$, places weight-norm growth inside the mechanism the bound describes.

4.6 The Experiment Grid

The sweep is organized into six *arms*, in the clinical-trial sense: one control condition, four that vary a single factor against it, and a hyperparameter search. Together they total 534 runs.

Arm	Configuration	Depths	Seeds	Runs
A	Unmitigated sweep: `{gcn, sage, gat}`	2-32	10	150
B	Mitigation ablation on GCN: `{residual}`, `{pairnorm}`, `{jk}`, `{pairnorm, residual}`	2-32	10	200
C	GCNII	2-32	10	50
D	Arm B's most effective mitigation, carried to `{sage, gat}`	2-32	10	100
E	Fidelity: GCN, `hiddenDim=16`	2	10	10
F	Hyperparameter search: GCN, `learningRate` x `dropout` x `weightDecay`	2	3	24
				534

Arm D is generated only after arm B is aggregated, its mitigation selected from arm B's outcome (Section 6.8); arm F's grid is `learningRate` in `{0.005, 0.01}`, `dropout` in `{0.5, 0.6}`, and `weightDecay` in `{5e-4, 1e-3}`.

The depth grid `{2, 4, 8, 16, 32}` is log-spaced because Cai and Wang's contraction bound (Cai and Wang 2020) predicts exponential energy decay in depth: log-spaced depths are evenly spaced on the log-energy axis the energy figures use. A candidate additional depth of 24 was rejected, since it buys resolution only where the curve has already collapsed and breaks that spacing.

The hyperparameters are tuned once, in arm F: GCN at depth 2, over the eight candidate configurations. The selected values are then frozen across every architecture and depth in arms A through E. Tuning per depth or per architecture would confound the depth effect the study measures with a hyperparameter effect. Section 6.2 reports the outcome, and its limitation: three seeds do not cleanly separate the eight candidates.

The published setups differ per architecture (GAT at learning rate 0.005 and dropout 0.6; GCNII at dropout 0.6 with a larger weight decay), so holding one GCN-tuned configuration fixed understates the architectures furthest from it, GAT foremost.

5 Explanation of the Source Code

> Drafting note, to be removed once resolved. This section is drafted directly from the source code, as the best available substitute for a defended walk-through. The outline this report follows states plainly that Section 5 should be written only for components that have passed a defense gate in the Claude.ai Project, since a subsection written about code nobody has walked through line by line is exactly the passage that cannot be defended on video or survive the course's zero-tolerance plagiarism review. That gate history is not visible from this session. Before submission, each subsection below needs to be checked against the actual gate record and either confirmed or reworked.

This section is a walkthrough of how the code expresses the design Section 4 describes, at a level where a reader with the repository open can follow along. It is not Section 4 restated and not a code listing. Detail is kept at the per-module level throughout: the key classes and functions in each module, the data flowing between them, and short excerpts only where an excerpt makes a point prose alone cannot.

5.1 Repository Map and Reading Order

5.1.1 Dependency Order

The build order and the dependency order coincide, and are the order to read the modules in:

Order	Module	Depends on	Notes
1	`data`	(none)	Root of the dependency graph
2	`models`	`data`, field names only	Declares the two protocols `mitigations` implements, without importing `mitigations`

Order	Module	Depends on	Notes
3	`metrics`	`data`; `models`, output shape only	A fixed instrument, not configuration-aware; does not depend on `train`, `mitigations`, or `experiments`
4	`train`	`models`, `metrics`	Owns the only optimizer in the repository
5	`mitigations`	The two protocols `models` declares	Sits after `train` so the null-object default lets the harness run and the baselines reproduce before any mitigation exists
6	`experiments`	Every other module	The only module that writes to disk
7	`viz`	`data`, labels only; the records `experiments` writes	Never imported by anything: the leaf of the graph and the only module whose output a reader sees directly

5.1.2 Entry Points

Three entry points sit outside this module graph:

- `src/run_sweep.py` executes the sweep against the six arms, in the order F, A, B, C, E, D.
- `src/generate_report_figures.py` regenerates every figure and table from `results/` alone.
- The pytest suite under `src/tests/` is run directly, rather than through either driver.

Neither `src/notebooks/` nor `src/tests.ipynb` is ever the source of truth for a component's implementation; both are for exploration and report-figure generation only.

5.2 Shared Interfaces in Code

5.2.1 The Model's Return Type

`GnnModel.Forward(x, edgeIndex) -> tuple[Tensor, list[Tensor]]` (`src/models/gnn_model.py`) is the one place this interface is expressed in code: it returns `(logits, layerEmbeddings)`, with `layerEmbeddings[0]` always the raw input tensor `x` and every subsequent entry the corresponding layer's post-activation, pre-dropout output.

5.2.2 The Results Record

The results record is assembled entirely inside `train.TrainRun` (`src/train/harness.py`) and written by exactly one caller, `experiments.RunSweep` (`src/experiments/runner.py`), via `_AtomicWriteJson`. `train` never touches the filesystem; `TrainRun` returns a plain `dict`, and a test can call it directly against an in-memory config with no filesystem fixture. This single-writer property is what keeps the filename convention (`experiments.runner.ResultPath`) in one place: nothing else in the repository constructs a `results/*.json` path independently, so the convention cannot drift between two call sites.

5.2.3 Runtime Assertions

Two runtime assertions enforce invariants the type system cannot: in `experiments.runner.RunOne`, `("jk" in config.mitigations) == (config.readout == "jk")` is checked before training begins, so a `RunConfig` in which the mitigation list and the readout disagree about Jumping Knowledge fails immediately rather than producing a record with an internally inconsistent `config` block; and in `metrics.oversmoothing.BuildAugmentedOperator`, the incoming `edgeIndex` is checked for self-loops and the constructor raises if any are found, so a caller that passed an already-augmented graph could not silently double-augment every node's degree.

5.2.4 Attached Embeddings

One more invariant belongs here rather than being left implicit: `layerEmbeddings` is returned attached to the autograd graph, and detaching happens only at the three capture sites in `train`, never inside `models`. Under Jumping Knowledge the readout consumes the whole embedding stack to produce the logits, so the training loss backpropagates through every entry in the list; detaching inside `Forward` would silently zero the gradient to every layer except the last, and JK would train incorrectly with no error raised anywhere.

5.3 Per-Module Walkthroughs

Module	File(s)	Owns	Does not own
`data`	`cora.py`	`LoadCora`, `AssertGraphInvariants`	Self-loop insertion, degree normalization
`models`	`gnn_model.py`, `architectures.py`, `protocols.py`	The layer loop, the four architecture subclasses	Mitigation logic (declares an interface for it only)
`metrics`	`oversmoothing.py`	`DirichletEnergy`, `MeanAverageDistance`, `SelectComparableBand`, `FitContractionSlope`	Training, mitigations, experiment configuration
`train`	`harness.py`	The optimizer, the training loop, result-record assembly	The filename convention, any file I/O
`mitigations`	`hooks.py`, `readouts.py`, `naming.py`	Three mechanism implementations, one naming helper	`GnnModel`, `train`, `experiments`
`experiments`	`grid.py`, `runner.py`	The declarative grid, model construction from a `RunConfig`, all filesystem writes	Any numerical computation

Module	File(s)	Owns	Does not own
`viz`	`aggregation.py`, `figures.py`, `tables.py`	Loading, tidying, aggregating, plotting, table export	Model construction

What follows for each module is the reasoning the table cannot carry: the non-obvious implementation choices and why each one exists.

5.3.1 `data`

Every invariant check (`num_nodes`, feature shape, class count, undirectedness, self-loop absence, mask sizes, pairwise mask disjointness) is a single vectorized tensor reduction (`.sum()`, boolean `&`, PyG's `contains_self_loops`); there is no Python loop over nodes anywhere in the module. The one non-obvious point: `AssertGraphInvariants` raises `ValueError` explicitly rather than using a bare `assert`, so the checks cannot be compiled out under `python -O`.

5.3.2 `models`

Mitigations attach to `GnnModel` by composition, not inheritance: they are constructed independently and injected at model construction as a list of `LayerHook` objects plus one `Readout` object, and `models` imports nothing from `mitigations`; both sides depend on two protocols declared inside `models` (the Appendix's class diagram gives both protocols' exact signatures). Inheritance was rejected because a mitigation is a modification applied to a model, not a kind of model: encoding residual, PairNorm, or Jumping Knowledge as an architecture subclass would need four mitigations times three architectures as classes before any combination is considered, and residual-plus-PairNorm together has no honest class name. Mitigation methods on the base class were rejected for a related reason: that moves the coupling up a level rather than removing it, since the base class would then have to know the full set of mitigations that exists and every new one would edit both the base class and its dispatch logic. The unmitigated baseline is a null object, not a branch: it is constructed with `layerHooks = []` and `readout = LastLayerReadout()`, and the layer loop contains no "are there mitigations?" test anywhere.

`GnnModel.__init__` builds `self.convs` as an `nn.ModuleList` by calling the subclass-supplied `BuildLayerConv` once per layer, computing each layer's input and output width from `numLayers`, `hiddenDim`, and whether the current layer is the final, logit-emitting one. `Forward` itself, reproduced below, is where the recording point from Section 4.2 is visible in code:

```
def Forward(self, x: Tensor, edgeIndex: Tensor) -> tuple[Tensor, list[Tensor]]:
    """Returns (logits [N, C], layerEmbeddings)."""
    h = x
    x0 = x
    layerEmbeddings: list[Tensor] = [x]
    for l, conv in enumerate(self.convs, start=1):
        hPrev = h
        h = conv(h, edgeIndex, x0)
        for hook in self.layerHooks:
            h = hook.Apply(h, hPrev, edgeIndex)
        isFinalAndLogits = (l == self.numLayers) and self.readout.FinalLayerIsLogits
```

```
        if not isFinalAndLogits:
            h = torch.relu(h)
        # tapping after activation, before dropout: the representation the
        # next layer actually receives, independent of the stochastic
        # dropout mask
        layerEmbeddings.append(h)
        if not isFinalAndLogits:
            h = self.dropoutLayer(h)
    logits = self.readout.Apply(layerEmbeddings)
    return logits, layerEmbeddings
```

(`src/models/gnn_model.py`, `GnnModel.Forward`, reproduced verbatim except for the outer method indentation.)

`GcniiModel.Forward` (in `architectures.py`) duplicates this loop rather than reusing the base class's, since its input and output projections do not fit the base class's per-layer construction: this is the one place in `models` where the same shape of loop is written twice, a direct consequence of `GCN2Conv`'s equal-width constraint. The non-obvious point worth a reader's attention: `_GatConvAdapter` divides the requested total output width by the head count before constructing `GATConv`, so that `hiddenDim` denotes total width across heads rather than per-head width, keeping GAT's capacity comparable to GCN's and GraphSAGE's at the same `hiddenDim` rather than inflated by the head count.

GCNII's projections are the other place `models` departs from the shared construction path. PyG's `GCN2Conv` requires its input and its initial-residual term at equal width, which none of the other three convolutions require, so `GcniiModel` cannot reuse the generic per-layer construction that lets `GcnModel/SageModel/GatModel` map `1433 -> hiddenDim` on layer 1 and `hiddenDim -> outDim` on the last layer inside the conv itself. `GcniiModel` instead bypasses `GnnModel.__init__` entirely (it calls `nn.Module.__init__` directly) and wraps exactly `numLayers` `GCN2Conv` hops at constant `hiddenDim` width, preceded by an uncounted `Linear(1433, hiddenDim)` input projection and, when the readout is `LastLayerReadout`, followed by an uncounted `Linear(hiddenDim, 7)` output projection that stands in for the final layer's activation. Neither projection counts toward `numLayers`, and the projected input is never written into `layerEmbeddings`: index 0 remains the raw input `X` for GCNII exactly as for every other architecture.

5.3.3 metrics

`DirichletEnergy` is one row-scale, one gather by edge index, one squared-difference row-norm, and one sum: `[2708, hiddenDim]` in, one Python `float` out, for every layer in a run. The numerical guards, and why each exists, are what this module turns on. `MeanAverageDistance` clamps the row-norm denominator to `1e-12` before normalizing, since ReLU produces exact all-zero rows at depth and an unclamped division would produce `nan` rather than a defined cosine distance; it separately clamps both of MAD's own reduction denominators (Eq. 3's per-row count and Eq. 4's final count) to return `0.0` rather than `NaN` when a count is zero, which is what a fully collapsed representation (every row identical, every pairwise distance zero) produces. Without these guards, the metrics this study's entire depth-32 result rests on would return `NaN` in precisely the regime the study is about, rather than the well-defined zero that regime should report. `FitContractionSlope` carries the third guard, the `1e-12` energy floor discussed in Section 4.5, needed because a genuinely collapsed configuration's per-dimension energy underflows to exact-zero in float32 and `log(0)` is undefined.

5.3.4 train

TrainRun is the sole public entry point most other code calls; EvaluateSplit, CaptureMetrics, and MacroF1 are its private-in-spirit helpers, reused directly by the per-epoch validation pass and by the three capture points. The training loop itself (the for epoch in range(1, maxEpochs + 1) block) is 20-odd lines: forward, cross-entropy on train_mask rows, backward, optimizer step, then an eval-mode validation pass whose loss decides whether bestState is updated and whether the patience counter resets. Vectorization here is inherited rather than original to this module: the forward and backward passes vectorize because models and PyG's message-passing kernels do, and MacroF1's own vectorization (a bincount over a combined class-pair index) is described in Section 4.4.

5.3.5 mitigations

ResidualHook and PairNormHook carry no learnable parameters and are plain Python objects rather than nn.Module subclasses; JkReadout does carry a Linear layer and is an nn.Module, since GnnModel stores readout as a direct attribute and PyTorch auto-registers it as a submodule, which is what lets that layer's parameters reach the optimizer alongside every convolution's. JkReadout.Apply asserts that every tensor in layerEmbeddings[1:] shares one width before pooling, so a misconfigured model would raise rather than silently broadcasting a shape mismatch into the pooling operation.

5.3.6 experiments

BuildGrid is six private _BuildArm* functions behind one dispatcher, each a small set of nested for loops over Python lists with no side effects: calling BuildGrid twice returns equal lists and touches no filesystem, which is what lets the grid's size be asserted directly in the test suite. SetSeed(config.seed) is called twice on the path from configuration to trained model (the Appendix's sequence diagram gives the exact call order). The first call is not redundant: weight initialization happens at model-construction time, which runs strictly before TrainRun's own seeding, so a run seeded only inside TrainRun would draw its initial weights from whatever random state happened to be ambient at the call site rather than from the configured seed. Without the earlier call, two runs sharing config.seed would produce different initial weights, which would break both the determinism the test suite asserts and the reproducibility claim the README makes.

RunSweep is idempotent, with atomic writes and exactly one writer: it skips any configuration whose result file already exists unless explicitly forced, records are written to a temporary path and renamed into place so a file is either complete or entirely absent, and experiments is the only module that ever writes to results/. At 534 runs over a single-machine, multi-day window, an interrupted sweep restarting from zero each time is the failure mode this design exists to avoid; a failing individual run is caught, written as a <runId>.failed.json marker alongside its error and traceback, and the sweep continues, so a genuine failure remains visible and distinguishable from a configuration that was simply never scheduled, rather than silently aborting the remaining several hundred runs.

5.3.7 `viz`

Aggregation is strictly separated from plotting: `LoadRecords` reads and validates every JSON record in a directory (raising on a malformed file rather than skipping it silently, so a partially written or corrupted record surfaces immediately rather than quietly shrinking a seed count), `BuildTable` flattens each record to one tidy row, and `Aggregate` groups by a set of columns and returns mean, standard deviation, and count per group; only after this does any `Plot*` function touch `matplotlib`. Mean and standard deviation are computed at read time and never written back to `results/`, so the records directory remains the single source of truth with no derived file beside it that could go stale. The comparable band is read from each record's stored `bandIndices`, never recomputed by the plotting code, mirroring the derivation rule in Section 4.5 from the consuming side: a plotting function that assumed the `LastLayerReadout` band (`1..L-1`) would silently truncate every Jumping Knowledge curve by exactly one layer, which is the failure this design choice exists to prevent, and it is checked directly in the viz test suite (Section 5.4). `BuildTable` and `Aggregate` are the two functions every `Plot*` function and every exported table pass through; no plotting function reads a raw JSON record directly except `PlotEnergyVsLayer` and `PlotLossCurves`, which need one specific run's full per-layer or per-epoch array rather than a cross-seed aggregate.

5.4 Testing

5.4.1 Test Categories

The pytest suite under `src/tests/` replaces the "runs top to bottom without errors" guarantee a single assembled notebook would otherwise have to provide informally. Four categories of test recur across the seven test modules: numerical correctness against a hand-computable or externally verifiable reference (for instance, `metrics`' dense-reference energy test, which checks the vectorized edge-list computation against a dense `trace(H^T * Laplacian * H)` form on a small synthetic graph, and `train`'s macro-F1 test against `sklearn.metrics.f1_score`); grid enumeration and uniqueness (`experiments`' grid-arithmetic and no-duplicate-paths tests); shape and interface assertions (`models`' band-width-homogeneity and final-entry-identity tests, conditioned on the readout); and qualitative gates, which check a directional property on real, trained output rather than an exact value (the smoke test that a 2-layer GCN reaches roughly 81 percent test accuracy; the depth-32 collapse gate described next).

5.4.2 Defects Caught by Testing

Two tests caught real defects before they reached a reported number, and both are worth naming as such rather than only as passing tests. The path-collision tests in `test_experiments.py` (`test_arm_e_and_f_collide_within_a_flat_directory` and `test_arm_f_still_collides_within_its_own_s` are what surfaced a filename collision between arms E/F and arm A, which the results filename convention (Section 4.6) now avoids by giving each its own subdirectory. Without these tests, arm E's ten fidelity runs and several of arm F's hyperparameter-search runs would have silently overwritten arm A's result files, corrupting the depth-sweep aggregate with no error anywhere. The collapse-floor tests in `test_metrics.py` (`test_collapse_floor_on_regular_graph` and `test_collapse_floor_on_cora_is_sqrt_degree_weighted`) are what surfaced that Cora's actual null space is the `sqrt(degree)` direction rather than the literal constant vector, a property of the symmetric-normalized Laplacian on an irregular graph that an assumption of "identical rows collapse to zero energy" would have gotten wrong on Cora specifically, even though it holds on a regular graph.

5.4.3 The Epoch-0 Gate

One test is gated on the epoch-0 capture rather than the checkpoint capture, and the reason is itself a result. `test_depth_qualitative_gate_at_epoch_zero` asserts that a 32-layer unmitigated GCN shows monotonically decreasing per-dimension energy at initialization, not after training: at the checkpoint, under this study's settled hyperparameters, that monotonic property does not hold (Section 6.5 reports why), so gating the test there would make it flaky or outright false under a correct implementation. This connects directly to Section 6.5's central finding: separating "the propagation operator collapses representations" from "training collapses representations differently" is not only a results-section distinction, it is a distinction the test suite itself had to make to stay meaningful.

5.5 What the Code Does Not Do

No per-layer gradient norm and no per-layer weight norm is persisted anywhere in the results schema, for any run. `TrainRun` calls `.backward()` every training step but never captures or stores the resulting gradient magnitudes, and no run's trained weights are checkpointed to disk once training completes. This is stated plainly here because Section 6 and Section 7 both depend on it: it is the reason the mechanism behind GCN's depth-32 early-layer energy collapse (Section 6.5) remains an inferred explanation rather than a confirmed one, and it is the single instrumentation gap Section 7.2 identifies as most directly closing that gap if added.

6 Results and Discussion

Every number in this section is read directly from `results/*.json` or computed from records there by `src/generate_report_figures.py`; none is transcribed from a conversation, a summary, or a remembered run. This section also distinguishes two kinds of claim throughout: a measured claim is read directly from recorded data and is stated flat; an inferred claim is a candidate explanation consistent with the measured data but not independently confirmed by it, and is phrased as *consistent with*, not *because*. This distinction is preserved throughout this section rather than allowed to blur, since conflating a measured result with an inferred one is the easiest way for a claim in this section to overreach what the data supports.

6.1 Reporting Conventions

Every result below is reported as mean and standard deviation across 10 seeds (3 seeds for the hyperparameter search, Section 4.6), on the standard 140/500/1000 public Cora split, unless stated otherwise.

Accuracy and macro-F1 (Section 3.19) are summarized by arithmetic mean; the energy ratio (Section 3.16) is summarized by geometric mean and range, and the two conventions are not interchangeable. Accuracy and macro-F1 are bounded, roughly symmetric quantities across seeds, for which an arithmetic mean is the ordinary and appropriate summary. The energy ratio `E_last/E_1` is a multiplicative, log-scale quantity that spans many orders of magnitude across seeds (26 orders of magnitude at depth 32, Section 6.5). An arithmetic mean of such a quantity is dominated by whichever single seed happens to be largest, to the point that "the mean of 10 seeds" can be arithmetically close to "one seed's value, divided by 10." Geometric mean is the summary statistic appropriate to a log-scale quantity, and it is used for every energy-ratio figure and table below; where a range is also reported, it is the minimum-to-maximum span across the 10 seeds, not a standard deviation, since a standard deviation of a quantity spanning 26 orders of magnitude is not a meaningful number.

Every energy (Section 3.13) or MAD (Section 3.14) figure and table below names which of the three capture points (Section 3.12) it reads: epoch 0 (before any gradient step), the selected checkpoint (the weights the reported test accuracy comes from), or the final epoch (the weights the training loop ended with, before any restore). A number or curve with no named capture has implicit, and therefore unverifiable, provenance; this report does not present one.

6.2 Baseline Reproduction

The fidelity arm confirms the baseline-reproduction claim: `hiddenDim=16` and `hiddenDim=64` give statistically indistinguishable test accuracy at depth 2. Arm E (`hiddenDim=16`, matching Kipf and Welling's published width) reaches mean test accuracy 81.60% (± 0.31); arm A's depth-2 GCN subset (`hiddenDim=64`, the width used throughout the rest of this sweep) reaches 81.49% (± 0.32). The difference, 0.11 points, is well within one standard deviation of either arm and not distinguishable given the observed seed variance. Both land close to Kipf and Welling's own published figure of approximately 81.5%.

Table 1: Depth-2 GCN test accuracy at the published width (`hiddenDim=16`, arm E) versus this study's sweep width (`hiddenDim=64`, arm A). Ten seeds each.

convType	hiddenDim	numLayers	testAccuracy_mean	testAccuracy_std	count
gcn	16	2	0.8160	0.0031	10
gcn	64	2	0.8149	0.0032	10

This licenses every subsequent deviation from the published width: since the width change from 16 to 64 does not materially move the depth-2 baseline, the capacity-parity reasoning for using 64 throughout the sweep (Section 4.2) costs nothing measurable at the one depth where a published number exists to check it against.

The hyperparameter search (arm F) selects a configuration that coincides exactly with Kipf and Welling's published one, but the search itself does not cleanly separate its eight candidates. Ranked by mean validation accuracy over 3 seeds:

Table 2: Arm F, all eight hyperparameter combinations, GCN at depth 2, ranked by mean validation accuracy over 3 seeds.

learningRate	dropout	weightDecay	valAccuracy_mean	valAccuracy_std
0.01	0.5	0.0005	0.8047	0.0042
0.01	0.5	0.001	0.8000	0.0020
0.005	0.5	0.001	0.7993	0.0031
0.01	0.6	0.0005	0.7993	0.0031
0.01	0.6	0.001	0.7987	0.0023
0.005	0.6	0.0005	0.7980	0.0020
0.005	0.6	0.001	0.7967	0.0012

learningRate	dropout	weightDecay	valAccuracy_mean	valAccuracy_std
0.005	0.5	0.0005	0.7933	0.0042

The winner (`learningRate=0.01, dropout=0.5, weightDecay=0.0005`) wins on the primary selection criterion (highest mean validation accuracy) without needing the tie-break rule: the runner-up sits 0.0047 below it, only marginally outside the winner's own one-seed-noise band of 0.0042. Read plainly, though, three seeds do not cleanly separate these eight configurations: all eight span a range of only 1.14 accuracy points, while individual per-configuration standard deviations (from only 3 seeds) run 0.12 to 0.42 points, comparable in scale to the total spread across all eight. What makes the winner a reasonable choice regardless of that closeness is that it is *also* the configuration closest to the published baseline, the search's own final fallback criterion, so the practical selection does not depend on trusting a ranking this close to noise: the fallback criterion alone would have selected the same configuration even if the primary ranking were pure noise.

6.3 Depth Sweep: Classification Performance

Test accuracy and macro-F1 against depth, for the unmitigated arm A sweep (three architectures) and arm C (GCNII, discussed on its own terms in Section 6.7):

Table 3: Test accuracy and macro-F1 versus depth, unmitigated GCN/GraphSAGE/GAT, ten seeds each. Macro-F1 computed directly from `results/*.json`, not previously tabulated by `src/viz`.

convType	depth	testAccuracy_mean	testAccuracy_std	macroF1_mean	macroF1_std
gcn	2	0.8149	0.0032	0.8032	0.0032
gcn	4	0.7368	0.0656	0.7381	0.0495
gcn	8	0.3349	0.1162	0.2864	0.1013
gcn	16	0.2352	0.0351	0.2104	0.0309
gcn	32	0.2407	0.0160	0.2102	0.0161
sage	2	0.8010	0.0042	0.7917	0.0040
sage	4	0.7172	0.0463	0.7086	0.0479
sage	8	0.2722	0.1078	0.0902	0.0734
sage	16	0.2850	0.0717	0.0627	0.0135
sage	32	0.2265	0.0983	0.0514	0.0189
gat	2	0.7890	0.0157	0.7780	0.0153
gat	4	0.7450	0.0593	0.7391	0.0641
gat	8	0.3228	0.1005	0.1563	0.1540
gat	16	0.2845	0.0727	0.0626	0.0137
gat	32	0.1966	0.0846	0.0459	0.0160

All three conventional architectures degrade with depth, and GCNII is the sole exception (Table 3, Figure 9; GCNII's own curve is deferred to Section 6.7). The degradation is not gradual across the full grid: all three drop 4.4 to 8.4 points from their depth-2 accuracy at depth 4, then drop sharply between depths 4 and 8, and remain low through depth 32.

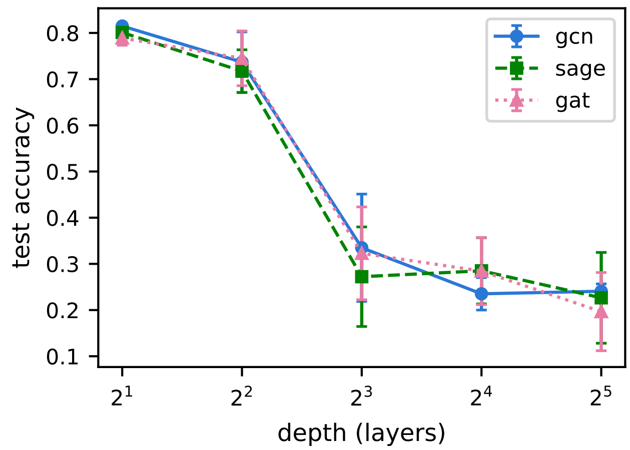


Figure 9: Test accuracy versus depth, unmitigated GCN, GraphSAGE, and GAT (arm A).

Macro-F1 collapses more sharply than accuracy for all three degrading architectures, most visibly for GraphSAGE and GAT. At depth 32, GAT's accuracy (19.66%) is roughly 5 points above the uniform-random floor for seven classes (14.3%, Section 3.19), while its macro-F1 (4.59%) is far below what either a uniform-random or majority-class predictor would produce on a balanced macro-averaged metric. This is *consistent with* predictions concentrating onto a small number of classes rather than spreading errors evenly across all seven. A per-class confusion matrix was not computed for this report, so that specific mechanism is not independently checked, only consistent with the size of the accuracy/macro-F1 gap.

The comparison against the majority-class floor (Section 3.19) is the sharpest single number in this depth sweep, and it holds for every seed, not only on average. The Cora test split's majority-class floor, computed directly from the test-mask label counts (`[130, 91, 144, 319, 149, 103, 64]`, summing to 1000), is 31.9%, not the approximately 30% figure sometimes quoted from memory. At depth 32, unmitigated GCN's mean test accuracy (24.07%) sits below this floor, and its full 10-seed range (21.2%-26.2%) sits below it entirely: even GCN's single best seed is 5.7 points short of a predictor that always guesses the majority class.

6.4 Structural Collapse at Initialization

At epoch 0 (before any gradient step, on the initialized weights only), GCN and GAT collapse cleanly and monotonically in both direction (MAD) and magnitude (energy ratio); GraphSAGE collapses in direction only. This is the initialized network as a whole: the propagation operator composed with each layer's randomly initialized, untrained weight matrices, not the propagation operator in isolation: the random weights' magnitudes are part of the composition and part of what drives the observed contraction, exactly as Cai and Wang's contraction bound has it (Section 3.18; `||W||` does not vanish merely because `W` is untrained).

Table 4: Epoch-0 (pre-training) MAD and energy ratio versus depth, unmitigated GCN/GraphSAGE/GAT.

convType	metric	depth 2	depth 4	depth 8	depth 16	depth 32
gcn	MAD	0.600	0.240	0.085	0.023	0.0045
gcn	energy ratio (`E_last/E_1`)	1.0 (trivial, single-point band)	3.14e-2	3.14e-4	2.60e-7	6.22e-13
gat	MAD	0.596	0.211	0.076	0.017	0.0041
gat	energy ratio (`E_last/E_1`)	1.0 (trivial, single-point band)	7.26e-2	2.73e-3	1.28e-5	1.25e-10
sage	MAD	0.083	0.0002	0.000	0.000	0.000
sage	energy ratio (`E_last/E_1`)	1.0 (trivial, single-point band)	18.3	19.8	20.3	19.3

6.4.1 Energy Ratio Decay and the Contraction Bound

GCN's energy ratio decays roughly 13 orders of magnitude by depth 32; GAT's decays similarly. Both are *consistent with* the qualitative shape Cai and Wang's bound predicts (repeated multiplicative contraction). This is explicitly not a claim of numerical agreement with the bound itself, since neither `||W||` nor the graph's spectral gap was computed and the measured decay rate was not checked against the bound's right-hand side.

6.4.2 GraphSAGE's Metric Dissociation

GraphSAGE's directional collapse (MAD) and its energy ratio dissociate, and the dissociation is explained by a measured geometric fact rather than left as a coincidence. GraphSAGE's MAD reaches near-zero already by depth 4, faster than either GCN or GAT, while its energy ratio *rises* from depth 2 to depth 4 and then plateaus around 19-20 at every depth tested, never decaying toward zero. Measured directly by singular-value decomposition at depth 16 (three seeds): GraphSAGE's representation matrix has top-singular-value energy fraction 1.0000, an exact rank-1 matrix, not merely dominated by one direction, and its top singular vector matches the constant direction with cosine 1.0000, versus cosine 0.9512 to the `sqrt(degree)` direction. GCN and GAT, by contrast, are only mostly rank-1 (energy fraction 0.93-0.98) and do not cleanly separate the two candidate directions. This connects directly to a fixed geometric property of the metric graph itself, defined in Section 3.3: the augmented, symmetric-normalized Laplacian's null space on Cora is the `sqrt(degree)` direction, not the literal constant vector, since Cora's degree distribution (2 to 169, Section 3.20) is far from regular. GraphSAGE collapses exactly onto the constant direction, which is provably *outside* that null space, so its Dirichlet energy cannot vanish regardless of how completely its representations collapse in direction. MAD is blind to which direction a collapse is onto (cosine similarity ignores the distinction); energy is not, and this is precisely why the two metrics disagree for GraphSAGE and not for GCN or GAT.

6.4.3 The Open Question: Why GraphSAGE Collapses First

GraphSAGE's MAD is already far below GCN's and GAT's at a single hop (depth 2), before any depth-driven stacking could plausibly apply: GraphSAGE MAD 0.083 versus GCN 0.600 and GAT 0.596. This was checked directly rather than only reported. Building one layer of each convolution type at matched random seeds on Cora's real graph reproduces the gap exactly (GraphSAGE 0.07-0.10 versus GCN/GAT 0.59-0.60 across three seeds). `root_weight` is not the cause: MAD stays low, in fact lower, with `root_weight=False` (0.03-0.05) than with the default `root_weight=True` (0.07-0.10), which rules out GraphSAGE's separate root term as an explanation. What remains open is narrower than the original question: why GraphSAGE specifically collapses onto the constant direction at a single hop, when GCN and GAT do not collapse this cleanly onto either candidate direction. Two verified-different candidates exist and neither has been isolated from the other: GraphSAGE's unweighted mean aggregation (against GCN's degree-normalized symmetric aggregation and GAT's attention weighting) interacting with Cora's skewed degree distribution; and a verified initialization difference (`GCNConv`'s and `GATConv`'s internal layers use Glorot initialization explicitly, while `SAGEConv`'s aggregation and root-transform weights fall back to PyTorch's `kaiming_uniform` default (Paszke et al. 2019), confirmed by reading each layer's source directly rather than assumed). Isolating either would require testing a degree-normalized-mean aggregation variant and a Glorot-initialized `SAGEConv` variant separately, neither of which has been done; this is recorded as an open question, not a finding, and is carried forward to Section 7.1.

6.5 The Trained State, and What It Does to the Collapse Signature

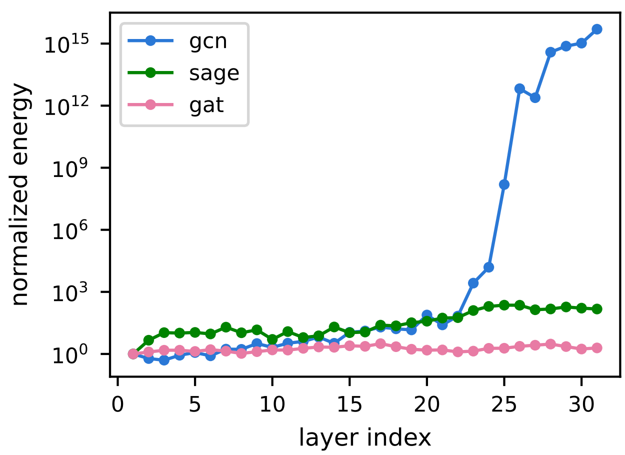


Figure 10: Normalized per-dimension energy versus layer index, checkpoint capture, depth 32, unmitigated GCN/GraphSAGE/GAT.

At the checkpoint (the weights the reported test accuracy comes from), GCN's collapse signature does not resemble its own epoch-0 signature, and reading it naively is easy to get backwards. Checkpoint MAD at the last band index, mean over 10 seeds:

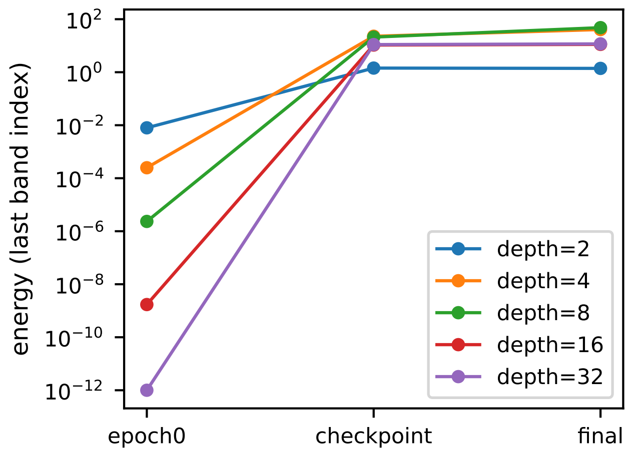


Figure 11: Per-dimension energy at the last comparable band index, across the three metric captures (epoch 0, checkpoint, final epoch) and across depth, unmitigated GCN. Geometric mean over ten seeds.

Table 5: Checkpoint (post-training) MAD versus depth, unmitigated GCN/GraphSAGE/GAT.

depth	GCN MAD	SAGE MAD	GAT MAD
2	0.240	0.262	0.286
4	0.264	0.291	0.302
8	0.290	0.058	0.115
16	0.176	0.000	0.000
32	0.183	0.000	0.000

6.5.1 Checkpoint MAD

GCN's checkpoint MAD is roughly flat and non-monotonic across the whole depth sweep, never approaching zero. SAGE and GAT's checkpoint MAD collapses to essentially zero by depth 16 and stays there, which is also a positive control on the metric itself: it fires cleanly where collapse is genuinely present (SAGE/GAT), which is what rules out metric insensitivity as the explanation for GCN not showing the same pattern.

6.5.2 Checkpoint Energy Ratio

GCN's checkpoint energy ratio grows explosively with depth (geometric mean, since this is exactly the log-scale quantity Section 6.1 flags):

Table 6: Checkpoint energy ratio versus depth, unmitigated GCN, geometric mean and range over 10 seeds.

depth	`E_last/E_1` geometric mean	range across 10 seeds
2	1.0 (trivial, single-point band)	1.0-1.0
4	26.6	16.0-43.6

depth	`E_last/E_1` geometric mean	range across 10 seeds
8	3.96e6	2.12e2-7.85e11
16	4.49e12	7.51e9-5.08e19
32	5.45e26	3.25e12-2.49e38

Reading this ratio as "nothing collapses at the checkpoint" is backwards, and an earlier pass at this analysis stated exactly that before it was corrected. The ratio is enormous because the *denominator* has collapsed, not because the numerator failed to shrink: `E_1` itself is approximately 1.4e-15 at depth 32, seed 0 (other seeds range down to 2.9e-38), already many orders of magnitude below a healthy energy value. Read correctly, the checkpoint state shows early-layer collapse together with late-layer growth (the early band carries essentially zero energy while the late band carries enormous energy), not an absence of collapse. This corrected reading is consistent with, not contradicted by, the early-layer collapse mechanism discussed below.

The largest individual per-seed ratio (2.49e38) sits within a factor of roughly 1.4 of float32's ceiling, but this specific number was never at risk of overflow: it is a division of two already-extracted scalar values computed at effectively double precision after being read out of JSON, not a float32 tensor operation. The check that matters is on the underlying `dirichletEnergy` tensor values themselves, computed in float32 before extraction: no `inf` or `nan` appears anywhere in any of the ten seeds' recorded energies, and the largest single raw energy value observed is approximately 2028, nowhere near a numerical ceiling. The ratio's enormous magnitude comes entirely from an astronomically small denominator, not from any value approaching overflow.

6.5.3 Per-Seed Consistency and the Seed 6 Exception

Cross-seed measurement confirms early-layer checkpoint collapse is a measured, not a one-run, property of unmitigated GCN at depth 32, with one exception reported as such, not smoothed into the average. Per-seed count of band layers (of 31) whose energy falls below the `1e-12` floor at the checkpoint:

Table 7: Checkpoint below-floor band layers, unmitigated GCN, depth 32, per seed.

seed	below-floor layers (of 31)	first layer at or above floor	`E_1`
0	22	23	1.4e-15
1	22	23	5.1e-20
2	26	27	7.5e-33
3	24	25	6.5e-24
4	25	25	2.9e-35
5	25	26	1.9e-31
6	0	1	3.3e-12
7	25	24	1.5e-26
8	25	22	4.2e-28
9	27	28	2.9e-38

Nine of ten seeds show the same shape: roughly the first 22-28 layers carry essentially zero energy, transitioning to non-negligible energy only in the last 4-9 layers, with the transition itself occasionally oscillating across the floor for three seeds (4, 7, 8) before settling rather than crossing it once cleanly. Seed 6 is a genuine exception, not folded into the pattern: no layer falls below the floor, and seed 6 also has both the highest test accuracy of the ten seeds (26.2%) and the shortest run (315 epochs versus 385-883 for the others), noted here as a correlation, not an explanation, and not investigated further (carried to Section 7.1).

6.5.4 The Inferred Mechanism

The mechanism behind this collapse (early-layer parameters starved of gradient signal while late-layer weights grow large and unconstrained) is *consistent with* the measured pattern above, but remains one-run inference, not a cross-seed measured fact. Terminal weight norms were inspected for exactly one run (`gcn_none_d32_s0`), post hoc, not as a persisted schema field and not as a gradient trace. No per-layer gradient norm exists anywhere in the recorded data for any run (Section 5.5), so this mechanism cannot be upgraded from *consistent with* to *measured* without new instrumentation (Section 7.2). It is deliberately not characterized further than "large and unconstrained": whether the late layers are fitting, or overfitting, the 140-node training set specifically is a separate, unmeasured claim. The schema has no training-accuracy field, and the measured test accuracy (24.07%, below the 31.9% majority floor) is, if anything, in tension with a clean overfitting story rather than supportive of one.

6.6 Separating Oversmoothing from Optimization Failure

Accuracy alone cannot distinguish "never trained" from "trained, then collapsed," and this schema was designed from the start to separate the two. At depth 32, both failure modes produce test accuracy near the majority-class floor; the results record's `trainingCurve` and `epoch0Metrics` fields exist specifically so the two are not conflated.

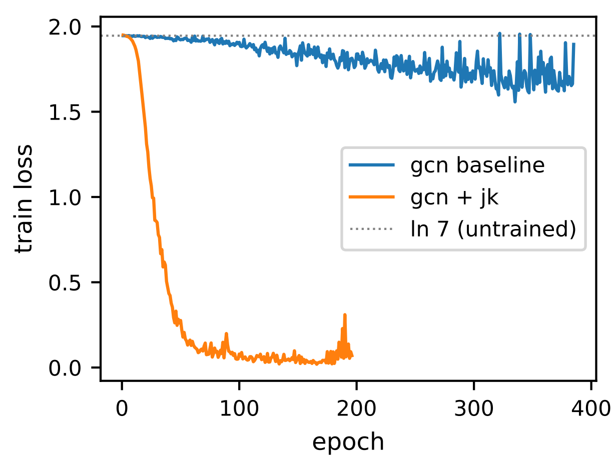


Figure 12: Training loss versus epoch, depth 32: unmitigated GCN baseline versus GCN with Jumping Knowledge.

6.6.1 Training-Loss and Stopping-Behavior Signals

The training-loss signal: GCN partially trains; GraphSAGE and GAT do not train at all. Across all 10 seeds, unmitigated GCN's training loss descends from `ln(7) = 1.9459` to a mean minimum of 1.57 (range 1.53-1.61), well above depth 2's well-fit final loss of approximately 0.23, but a clear, non-trivial descent. GraphSAGE and GAT's training loss stays flat at `ln(7)` (mean minimum 1.934 and 1.939 respectively, a movement of 0.01-0.02 nats, indistinguishable from optimization noise).

The independent stopping-behavior signal points the same way, and it is independent precisely because it falls out of the early-stopping mechanism itself rather than out of any analysis of the loss values or the metrics. GCN's runs last 315-883 epochs before `patience=100` exhausts, long enough that validation loss kept improving somewhere along the way. GraphSAGE and GAT's runs exhaust patience almost immediately (102-168 and 101-121 epochs respectively), consistent with validation loss never improving past its epoch-0 value.

6.6.2 Three Converging Signals

Three separately measured quantities converge on the same conclusion for GraphSAGE and GAT, none computed from either of the others, which is what makes this convergence rather than circularity: the flat training loss, the near-immediate exhaustion of patience, and the checkpoint's near-total MAD collapse (Section 6.5, essentially identical to the epoch-0 collapse in Section 6.4). Each is measured directly from the record; none is derived algebraically from either of the other two. That the three agree is read as *consistent with* training essentially never moving GraphSAGE's and GAT's weights at depth 32. This is stated as consistent with, not as independently confirmed beyond what the three measured signals jointly support, since no fourth, independent measurement (a gradient trace, for instance) was taken to confirm it directly. ### The Resulting Taxonomy

The resulting taxonomy is three distinct depth-32 failure modes across the three conventional architectures, not one shared failure: GCN partially trains and reaches a distinct, non-collapsed, non-recovered accuracy plateau below the majority-class floor; GraphSAGE and GAT do not train at all, and their checkpoint state is close to their own epoch-0 state by default, not because training reproduced it. An earlier draft of this analysis generalized GCN's partial-training pattern to all three architectures without checking it against GraphSAGE or GAT specifically; that generalization was false as stated, and the corrected, narrower taxonomy above replaces it rather than softening it.

6.6.3 The Positive Control

The positive control closes the loop on metric insensitivity as an alternative explanation. Because both metrics fire cleanly and collapse to essentially zero for GraphSAGE and GAT (Section 6.5), the same metrics' failure to show a comparable collapse for GCN's checkpoint MAD (Section 6.5) cannot be attributed to the metrics being insensitive in general. They are demonstrably sensitive to genuine collapse elsewhere in the same experiment, which is what licenses reading GCN's non-collapsing checkpoint MAD as a real property of GCN's trained state rather than as an artifact of the instrument.

6.7 The Architecture That Does Not Degrade

GCNII's test accuracy rises monotonically with depth, the opposite direction from GCN, GraphSAGE, and GAT:

Table 8: GCNII test accuracy versus depth, ten seeds (matches `tables/accuracy_vs_depth.md`).

depth	testAccuracy_mean	testAccuracy_std
2	0.7845	0.0179
4	0.7581	0.0482
8	0.7766	0.0163
16	0.8016	0.0104
32	0.8326	0.0044

Depth 32 is GCNII's best result and has the tightest cross-seed spread of any depth tested (0.44 accuracy points, versus 1-5 points at every other depth in this table).

GCNII's checkpoint contraction slope (Section 3.17) trends toward and past zero with depth, the opposite of GCN's explosive growth, rather than collapsing:

Table 9: GCNII checkpoint contraction slope versus depth, ten seeds. Depth 2 omitted (single-point band, `nan`).

depth	checkpointContractionS-lope_mean	checkpointContractionS-lope_std
4	+0.383	0.063
8	+0.169	0.024
16	-0.037	0.010
32	+0.012	0.003

At depth 16 the mean slope is slightly negative, a genuine sign flip, not noise around zero given the standard deviation of 0.010, and at every depth tested the slope stays within roughly ± 0.4, in sharp contrast to GCN's checkpoint slope, which grows into the tens by depth 8 and beyond under the same floored fitting procedure.

GCNII's identity-mapping and initial-residual terms are architecturally designed to prevent both the vanishing-gradient dynamic Section 6.5 infers for GCN and the unconstrained late-layer growth that produces it; the near-zero, non-exploding slope and the monotonically improving accuracy are *consistent with* that design succeeding at what it is meant to do. This has not been checked against GCNII's own per-layer weight norms or gradients the way GCN's mechanism was informally inspected in Section 6.5 (that inspection was not done for GCNII either), so this remains a plausible reading of the accuracy/slope pattern rather than a confirmed mechanism.

6.8 Mitigation Ablation

The full ablation at depth 32, all five arms including the unmitigated baseline:

Table 10: Mitigation ablation on GCN, depth 32, ten seeds each.

arm	testAccuracy_mean	testAccuracy_std	min	max
none (baseline)	24.07%	1.60	21.2%	26.2%
residual	19.25%	8.97	10.3%	32.1%

arm	testAccuracy_mean	testAccuracy_std	min	max
pairnorm	58.07%	7.10	47.6%	65.6%
jk	73.13%	3.43	68.0%	78.2%
pairnorm+residual	24.36%	9.79	13.5%	41.0%

6.8.1 The Full Ablation

Jumping Knowledge is the clear winner, roughly 15 points ahead of the next best mitigation and with the tightest cross-seed spread of any mitigated arm, the basis on which it was selected to carry forward to GraphSAGE and GAT as arm D. PairNorm helps substantially (58.07% against the 24.07% baseline) but well short of JK. Residual alone does not clearly beat the unmitigated baseline at depth 32: its mean (19.25%) sits below baseline's (24.07%), though its standard deviation (8.97) is large enough, and its maximum (32.1%) high enough, that this is not a confident "residual hurts" claim on ten seeds alone. Combining PairNorm with residual is worse than PairNorm by itself (24.36% versus 58.07%), a negative interaction between the two mechanisms at this depth, closer to the residual-alone and baseline numbers than to PairNorm's, rather than any combination of their individual benefits.

6.8.2 The Residual Prediction

A design decision anticipated the residual result before it was measured. Residual was designed, per Kipf and Welling's own Appendix B protocol, as a no-projection identity hook that cannot fire on the width-changing boundary layers (layer 1 and, under `LastLayerReadout`, the last layer), and the design note recorded at that time flagged explicitly that if the residual arm showed no benefit at depth, the absent layer-1 residual would be a candidate explanation. That candidate has not itself been checked (it would require a layer-1-projection diagnostic outside the current ablation), so this is the design's prediction confirmed as *worth investigating*, not confirmed as *correct*.

6.8.3 Unexplained Underperformance

No mechanism is proposed here for why residual underperforms baseline, or why PairNorm+residual underperforms PairNorm alone, at depth 32. Both are real, measured patterns in the aggregate accuracy numbers, but explaining them would need the same per-run scrutiny given to the unmitigated baseline in Section 6.3, Section 6.4, and Section 6.5 (training curves, checkpoint MAD and energy, informal weight inspection), and that scrutiny has not been given to any mitigated arm. This is recorded as an open question worth that scrutiny, not as a claim that a reason is already known (carried to Section 7.1).

6.8.4 Generalizing to Other Architectures

Whether the winning mitigation generalizes to the other architectures is answered by construction, not left open: Jumping Knowledge is exactly the mitigation carried to GraphSAGE and GAT in arm D, selected on this depth-32 comparison per the study's own selection rule (highest mean test accuracy at depth 32, not averaged across all five depths, since the ablation's purpose is restoring deep performance specifically). Arm D's own results are a direct extension of this table and are not restated separately here.

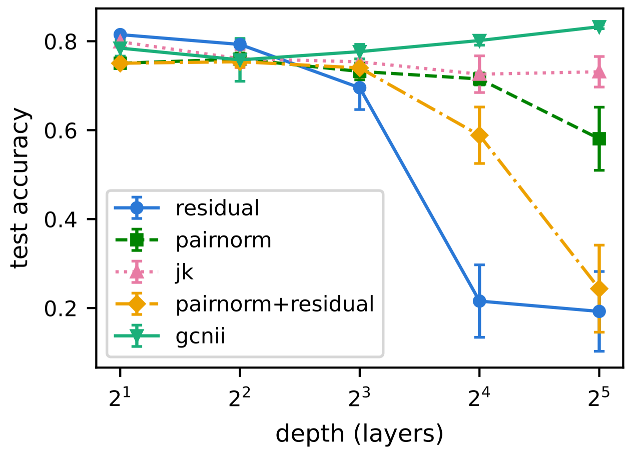


Figure 13: Test accuracy versus depth, GCN mitigation ablation (arm B) plus GCNII (arm C).

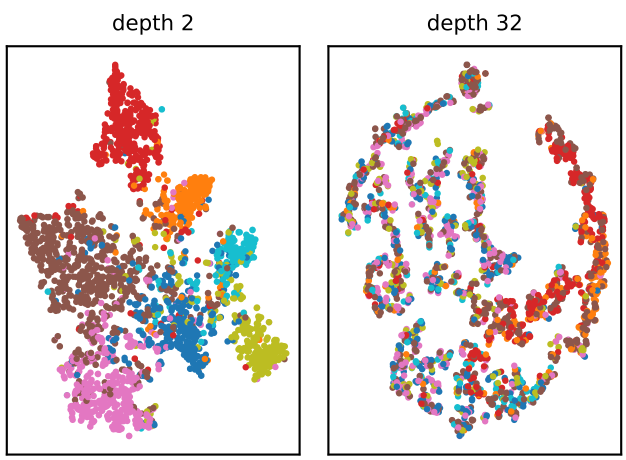


Figure 14: t-SNE projection of the checkpoint-capture layer embeddings, unmitigated GCN, seed 0, depth 2 versus depth 32, colored by class label. Fitted separately per panel; `random_state=0`, `perplexity=30`.

6.9 Qualitative Embedding Analysis

This projection illustrates collapse; it does not measure it. Dirichlet energy and MAD (Section 6.4 and Section 6.5) measure collapse directly and are the quantities this report's claims rest on; Figure 14 is labeled qualitative here deliberately, since treating a two-dimensional projection as a quantitative measurement would be an overclaim. The depth-2 panel is expected to show visibly separated class clusters, consistent with the shallow model's 81.49% test accuracy; the depth-32 panel is expected to show the classes substantially intermixed, consistent with the checkpoint MAD and energy patterns reported in Section 6.5 for unmitigated GCN at that depth. Whichever pattern the rendered figure in fact shows should be discussed here against what Section 6.4 and Section 6.5 report numerically, rather than assumed to agree with it. The two are independent views of the same run, and where they disagree that disagreement is itself worth a sentence.

The projection is fit separately for the shallow and deep panels: a shared fit across both would impose one neighborhood structure on both and manufacture the very difference the figure exists to illustrate.

Ten of the 534 configurations additionally persist their checkpoint-capture layer embeddings to `results/embeddings/*.pt`: GCN, seed 0, at depths 2 and 32, for each of the five mitigation arms (the unmitigated baseline plus arm B's four combinations). These are existing arm A and arm B configurations with a flag set, not additional runs, and exist because this analysis has no other source for the raw `[2708, 64]` embedding tensors; every other run writes only the reduced scalars `metrics` produces.

6.10 Threats to Validity and Limitations

Each deviation this study makes from a published or proposed setup has a consequence. They are enumerated here together with the direction of each effect, and whether it could have manufactured a finding or only understated one.

- Activation and dropout are uniform across all four architectures (ReLU, dropout 0.5), where GAT publishes ELU and dropout 0.6. Uniformity is what lets the recording position of Section 4.2 mean the same thing across architectures: ELU is not non-expansive with respect to Dirichlet energy the way ReLU is, so a mixed-activation study would attribute part of GAT's energy curve to its nonlinearity rather than to its aggregation. The cost is that GAT is not run at its published activation.
- Weight decay is applied uniformly to every layer, where Kipf and Welling apply it to the first layer only (Section 4.4). At depth 32, first-layer-only decay would regularize roughly a thirtieth of the parameters, making regularization strength an implicit function of depth. Uniform decay removes that dependence, at the cost of departing from the published two-layer setup.
- Checkpoint selection uses validation loss, where the submitted proposal specified validation accuracy (Section 4.4). Both quantities are recorded at every epoch in each run's `trainingCurve`, so the accuracy-selected alternative remains derivable from the stored records.
- GCNII runs a hybrid configuration: its architectural hyperparameters (`alpha=0.1`, `theta=0.5`) are kept from the paper, while its training hyperparameters come from this study's shared configuration rather than the paper's own. This is not a full reproduction of the published GCNII result and is not presented as one.

- The metric graph does not match GAT's own propagation graph (Section 4.5). The fitted contraction rate for GAT describes decay with respect to Cora's fixed structure, not GAT's attention-weighted one. This could either understate or overstate GAT's true propagation-relative decay rate, in a direction not established here, since GAT's attention weights were not compared against the fixed metric graph directly.
- The energy floor (`1e-12`) understates the magnitude of change at depth (Section 4.5). Quantified directly for the unmitigated depth-32 GCN checkpoint: the floor binds on 22 of 31 band layers, and the reported slope (`+0.8622`) is shallower than the unfloored value (`+1.1435`). This can only pull a reported effect toward zero, never manufacture one that is not present.
- Mitigations were run at untuned, fixed configurations (PairNorm's scale; Jumping Knowledge's aggregation mode). This can only understate their measured effectiveness relative to a tuned variant, never overstate it.
- One shared hyperparameter configuration was used across four architectures with differing published setups. GAT and GCNII are the architectures furthest from the shared configuration and are the ones most likely to be understated by it.
- The residual mechanism cannot fire on the boundary layers (layer 1 and, under `LastLayerReadout`, the last layer), by design (Section 4.3). This is a candidate, unverified explanation for residual's weak depth-32 result (Section 6.8), not a confirmed one.
- GCNII carries two uncounted projection layers the other three architectures do not (Section 4.2), which can only overstate GCNII's capacity relative to its peers, in GCNII's favor, at every depth.
- Ten seeds is the standard count in this study; three seeds is not enough, specifically in the hyperparameter-search arm (Section 6.2), where the eight candidate configurations' spread is comparable to individual per-configuration seed noise.
- Single dataset, single split. Every result in this section is specific to Cora's public 140/500/1000 split; nothing here establishes that the depth-failure taxonomy, the mitigation ranking, or the specific numbers generalize to another citation network, another homophily regime, or another split protocol.
- Findings that remain inferred rather than measured, named as such, with what would settle them: the parameter-level mechanism behind GCN's early-layer checkpoint collapse (Section 6.5; per-layer gradient norms would settle it, Section 7.2); the reason GraphSAGE and GAT's response to Jumping Knowledge or PairNorm differs from GCN's, since arm D's own per-run scrutiny was not given the same depth as the unmitigated baseline's; and GCNII's own weight/gradient mechanism (Section 6.7), for the same reason.
- A "fully collapsed" representation under this study's metrics is not literally identical across nodes. Section 6.4 establishes that GCN and GAT's epoch-0 collapse is not onto the literal constant vector but onto (or near) the augmented Laplacian's actual null space, `sqrt(degree)`-weighted on Cora's irregular graph; a reader should not take "collapsed" in this report to mean "every node's representation is bitwise identical."

6.11 Synthesis

The proposal anticipated progressive collapse with depth as a shared mechanism across architectures; the record in this section supports a narrower and more precise claim, and that narrowing is itself the contribution. Depth degrades classification performance for GCN, GraphSAGE, and GAT alike, but not through one shared failure mode: GraphSAGE and GAT at depth 32 do not train in any meaningful sense (Section 6.6), and their depth-32 representational state is close to their own untrained, initialization-time state rather than a product of training-induced collapse. GCN alone both trains partially and develops a distinct signature, early-layer energy collapse paired with late-layer energy growth (Section 6.5), that is neither present at initialization nor a simple continuation of it. GCNII, architecturally designed against exactly the mechanism inferred for GCN, degrades in neither sense and instead improves with depth (Section 6.7). Jumping Knowledge, not residual connections, is this study's most effective tested countermeasure for the architecture it was evaluated on (Section 6.8). The real-world applications named in Section 1, recommendation systems, social-network analysis, and drug discovery, differ from Cora in ways that plausibly favor one of the three failure modes above over the others, though none of the three is tested directly on any of them. Recommendation and social-network graphs built from user or member interactions typically have degree distributions far more skewed than Cora's own (Section 3.20), with a small number of hub users or popular items accumulating a disproportionate share of edges, and are frequently too large for full-batch training, so GraphSAGE's neighbor-sampling design, not the full-batch aggregation this study measures, is the form in which GraphSAGE is actually deployed at that scale. This study's finding that unmitigated GraphSAGE does not train at depth 32 on full-batch Cora is therefore not informative about GraphSAGE under sampling; the two settings differ in exactly the respect (which neighbors enter each layer's aggregation) this study holds fixed. Molecular graphs used for drug discovery are structurally the least similar of the three applications: individual molecules are small, low-diameter, and close to regular, each atom bonds to a handful of neighbors rather than to the hub structure that makes Cora's own null space diverge from a constant vector (Section 3.3), so the depth-32 failure modes measured here are unlikely to be the binding constraint, and shallow-to-moderate depth is already closer to what molecular property prediction typically uses. Which failure mode is most relevant to a given application is therefore a question of how closely that application's graph resembles Cora's degree distribution and depth requirement, not one this single-dataset design can settle. What this study's diagnostic apparatus offers to these domains is not a prediction of which architecture fails where, but a way of asking the same question its three capture points ask here, whether a deep model failed to train or trained and then collapsed, on the application's own graph.

7 Recommendations for Future Work

Every recommendation below is drawn from an open question Section 6 itself produced or from an instrumentation gap Section 5.5 names, rather than from a generic list of directions a depth-and-oversmoothing study could take.

7.1 Questions This Study Raised and Did Not Answer

GraphSAGE's single-hop collapse onto the constant direction is the most concrete open question this study produced. Section 6.4 narrows it to two verified-different, unisolated candidates: GraphSAGE's unweighted mean aggregation interacting with Cora's skewed degree distribution, and its `kaiming_uniform` initialization where GCN and GAT use Glorot. Two experiments would isolate them, run separately: a degree-normalized-mean aggregation variant holding initialization fixed, and a Glorot-initialized `SAGEConv` holding the aggregation fixed. Neither has been run.

The mitigated arms received none of the per-run scrutiny the unmitigated baseline received. Section 6.8's negative interaction between PairNorm and residual (worse combined than PairNorm alone) is measured but unexplained; the same training-curve, checkpoint-metric, and weight-inspection analysis given to the unmitigated baseline in Section 6.3, Section 6.4, and Section 6.5 has not been given to any of the four mitigated arms.

Seed 6's exception to GCN's early-layer collapse pattern (Section 6.5) is noted as a correlation, not investigated. It is the only one of ten seeds whose depth-32 checkpoint shows no band layer below the energy floor, and it also has both the highest test accuracy and the shortest training run of the ten; whether these three facts share a cause has not been examined beyond noting they co-occur.

7.2 Instrumentation Gaps

Gap	What it would settle	Current status
Per-layer gradient norms	Whether early-layer parameters receive vanishing gradient signal during training, converting Section 6.5's mechanism claim from inferred to confirmed	Not persisted; Section 6.5 rests on one run's terminal weight norms, inspected post hoc
Per-layer weight norms across the full sweep	Whether that single-run weight-norm observation is a cross-seed, cross-architecture pattern	Not persisted
Dense-capture trajectories over training	A continuous curve in place of the three-point epoch-0/checkpoint/final picture	The results schema reserves a nullable `trajectory` field, deliberately deferred rather than omitted (Section 4.4's capture-cost reasoning)
A training-accuracy field	Whether GCN's deep late layers are fitting, or overfitting, the 140-node training set	Does not exist in the results schema

7.3 Extensions to the Experimental Design

Extension	What it would show	Why not done here
Finer depth resolution between 4 and 8	Where within that interval the accuracy transition is sharpest	The log-spaced grid (`2, 4, 8, 16, 32`) only resolves that all three conventional architectures degrade somewhere in the interval

Extension	What it would show	Why not done here
Per-architecture hyperparameter tuning	Each architecture's own best achievable depth-32 performance	A different question from this study's; the shared, GCN-tuned configuration is deliberately held fixed so the depth comparison is not confounded by a hyperparameter effect (Section 4.6)
Additional datasets differing in homophily and degree distribution	Whether the `sqrt(degree)` null-space result (Section 6.4) is Cora-specific or general	Cora's degree range (2 to 169) is what makes the two candidate null-space directions distinguishable in the first place; a more regular graph would collapse that distinction
Decomposing contraction into aggregation-driven and nonlinearity-driven parts	How much of a layer's contraction comes from the aggregation operator versus ReLU's own non-expansiveness	The current recording point (post-activation, pre-dropout, Section 4.2) captures one combined tensor per layer, not both separately

7.4 Scale and Noise

Section 2.7 sets aside two properties of real deployments that Cora's controlled design does not test directly: scale, a graph too large for full-batch computation, and noise, unreliable or missing edges. This study's depth-failure taxonomy and diagnostic apparatus bear on both without having been run on either.

On scale, sampling changes what depth means operationally. GraphSAGE-style neighbor sampling turns an L-layer receptive field into a sampled approximation of the full L-hop neighborhood rather than an exact one. This study's finding that unmitigated GraphSAGE does not train at depth 32 on full-batch Cora is therefore not informative about the same architecture under sampling: sampling changes both the aggregation statistics at each layer and, from one training step to the next, which neighbors a node actually sees. Dirichlet energy and MAD transfer unmodified in definition, since both are computed on whatever propagation graph and node set a run supplies. What does not transfer without modification is the fixed metric graph this study uses for cross-architecture comparability (Section 3.22): sampling induces a different subgraph at each mini-batch, so applying the metrics under sampling first requires deciding whether energy and MAD are computed on the sampled subgraph or on a fixed full graph held aside for evaluation only. On noise, the same distinction this study draws between representational collapse and training failure stays relevant. A model trained on a graph with spurious or missing edges can lose accuracy for either of two reasons: noise degraded the label information available in an otherwise still-diverse representation, or noise accelerated collapse by adding uninformative connections into every layer's aggregation. Accuracy alone does not distinguish the two, for the same reason it does not distinguish them on Cora (Section 6.6). The three capture points used throughout this study would show whether a noisy graph's effect on representations is already present before training, a structural property of the noise itself, or develops only once training begins.

The project description's request for a proposed improvement is answered here from this study's own mitigation results rather than from an untested architecture. PairNorm is the only tested mitigation that controls representation dispersion directly, through normalization. Residual connections and GCNII's initial-residual term instead preserve an earlier representation through an identity path; PairNorm also adds no learned parameters, which keeps its cost independent of graph size. This study's contraction-slope results show that the rate of representation change is not constant across depth or architecture, which motivates scaling PairNorm's normalization strength to depth rather than applying it at one fixed scale throughout a network, as this study does. This proposal is not evaluated in this report; it is consistent with, not confirmed by, the mitigation ablation in Section 6.8, and is offered as the direction the measurements most directly support. ## Transfer to Physics-Informed Graph Learning

Section 1 motivates part of this study through physics-informed graph neural networks, where a physical state is propagated across a discretized domain and depth is set by the geometry of that domain rather than chosen freely. The framing committed to in the proposal is kept honest here: Cora is small and tightly clustered. This study's specific numbers, the majority-class comparison, the mitigation ranking, the depth at which each architecture's accuracy transitions, are not expected to transfer to a much larger mesh-structured graph with a different degree distribution and a different notion of locality.

What is expected to transfer is the diagnostic methodology:

- The three capture points (initialization, selected checkpoint, final epoch).
- A comparable band of layers derived from the model's own tensor widths rather than assumed in advance.
- Dirichlet energy and MAD reported as a pair rather than either alone.
- The practice of separating a trained state from an untrained one before attributing a failure to oversmoothing.

The qualitative behavior of the mitigations tested here, that a normalization-based intervention controls dispersion directly while an identity-path intervention preserves an earlier representation instead, is also a candidate for transfer, since neither mechanism is specific to Cora's graph. Overclaiming transfer beyond this would undo the care taken throughout Section 6 to keep measured and inferred claims separate: this study did not run a physics-informed model, and nothing here confirms that the same architecture-dependent taxonomy found for GCN, GraphSAGE, and GAT on Cora reproduces on a mesh graph. It is offered as the next test the diagnostic apparatus built here is designed to support.

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Appendix A: System Diagrams

These two diagrams are hand-drawn TikZ, not a UML-specific package: `pgf-umlcd`/`pgf-umlsd` exist and were tested, but fail to render class-title text under this document's LuaLaTeX engine (they work under `pdflatex`; switching engines this late over two figures was not worth it). Neither diagram adds a claim the prose in Section 5 does not already make; each is a visual index into the code structure that section already describes in words.

Figure 15, the class diagram, follows on the next page, rotated to landscape for its width. It lays out `GnnModel` and its four architecture subclasses (`GcnModel`, `SageModel`, `GatModel`, `GcniiModel`), the two composition seams the base class exposes, `LayerHook` and `Readout`, and the concrete classes realizing each: `ResidualHook` and `PairNormHook` on the mitigation side, `LastLayerReadout` and `JkReadout` on the readout side. The dashed open-triangle arrows read as "realizes an interface"; the solid open-triangle arrows read as "inherits from"; the dashed "uses" arrows mark where `GnnModel`'s constructor wires a hook or a readout into the forward pass without the base class knowing its concrete type, the dependency-injection seam Section 5.3's `mitigations` entry describes in prose.

Figure 16, the sequence diagram, follows immediately after it. It traces one call to `RunOne`: seeding twice, once for weight initialization and once to isolate training-loop randomness such as dropout (Section 5.3's `experiments` entry), the epoch loop that alternates a forward pass with the validation-loss check driving early stopping, and the three `CaptureMetrics` calls, at epoch 0, at the final epoch before the best-checkpoint weights are restored, and at that restored checkpoint itself, that Section 3.12 defines and Section 4.4 explains the ordering of. The diagram makes visible a detail easy to miss reading the code sequentially: the final-epoch capture happens on the weights the training loop ended with, before `load_state_dict` restores the best-validation checkpoint, not after.

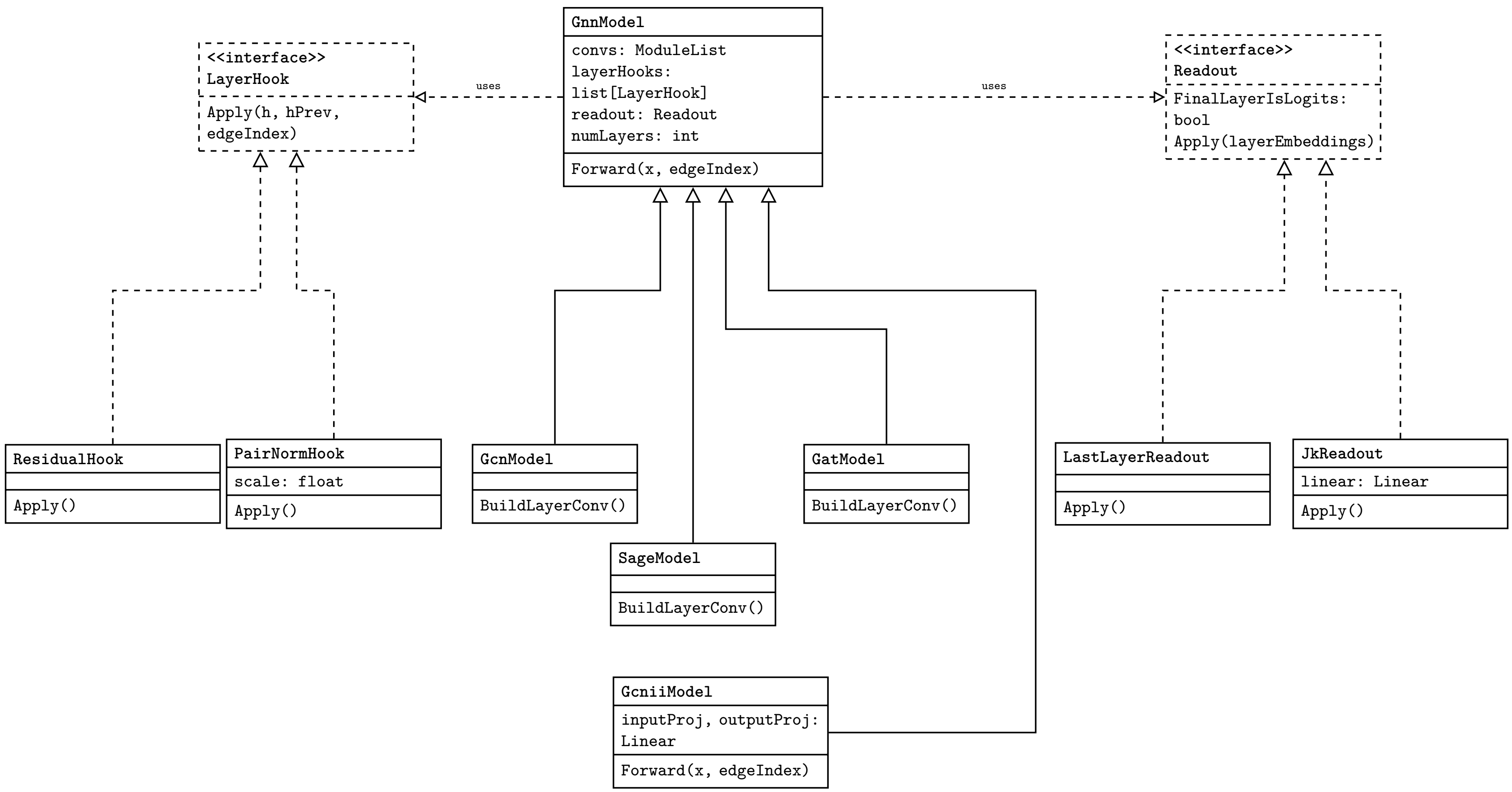

Figure 15: `GnnModel` and its four architecture subclasses (Section 5.3's `models` entry), the `LayerHook`/`Readout` composition seam `mitigations` implements against (Section 5.3's `mitigations` entry), and the two tested realizations of each protocol.

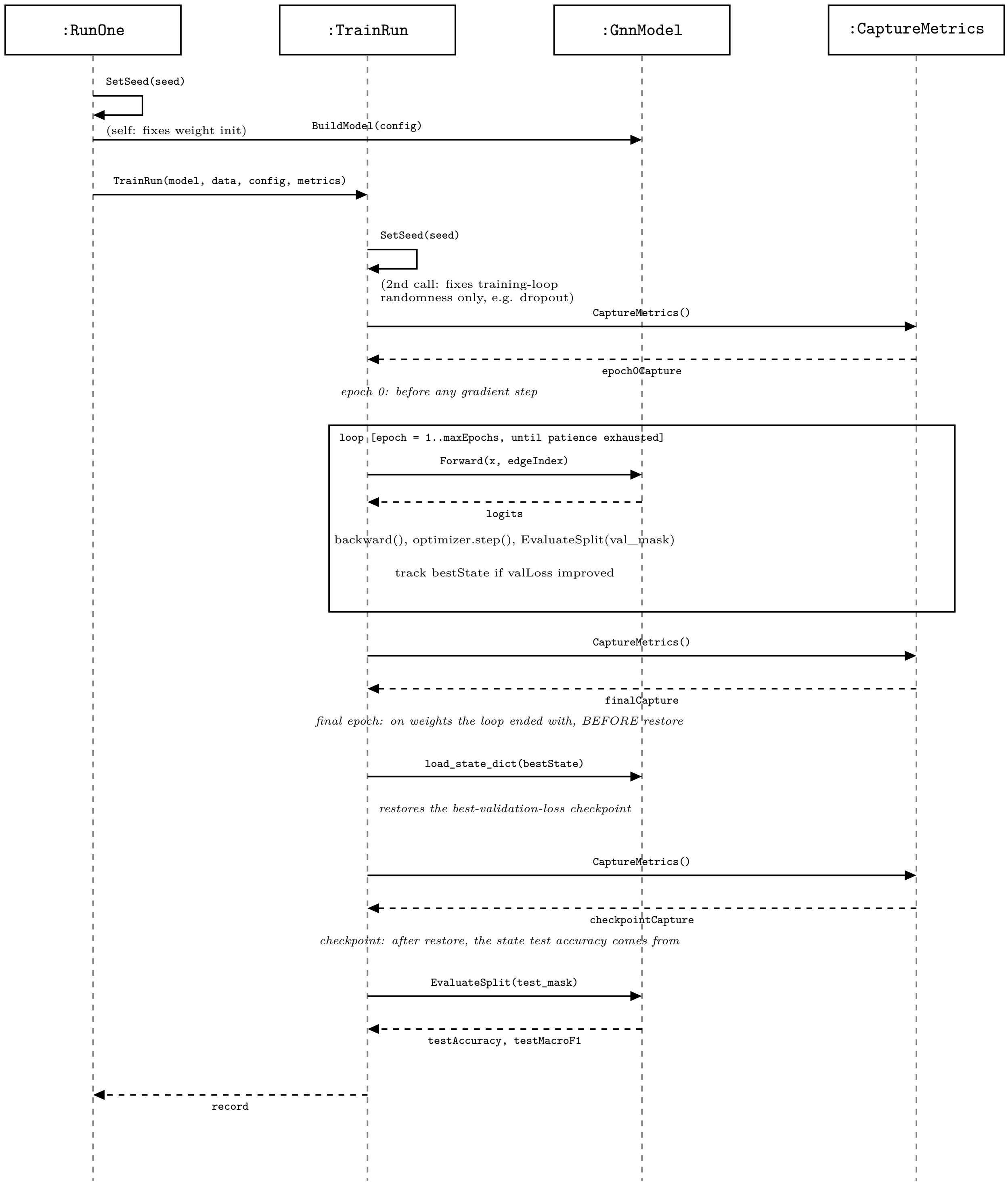

Figure 16: The call sequence for one run, from `RunOne` through the three metric captures (Section 3.12), showing the two `SetSeed` calls (Section 5.3's `experiments` entry) and that the final-epoch capture happens before, not after, the checkpoint restore (Section 4.4).

Appendix B: Full Hyperparameter Search Grid

Every table in Appendices B-D is generated directly by `src/generate_report_figures.py` from the real sweep in `results/`, not transcribed by hand; regenerating it after deleting `tables/` reproduces these exact rows. Each is the per-seed data underlying an aggregate already reported in the main text.

The 24 individual runs (8 hyperparameter combinations x 3 seeds) behind the aggregate in Table 2 (Section 6.2).

Table 11: Arm F, per-seed validation accuracy and loss, all eight hyperparameter combinations.

learningRate	dropout	weightDecay	seed	valAccuracy	valLoss
0.0050	0.5000	0.0005	0	0.7920	0.6987
0.0050	0.5000	0.0005	1	0.7980	0.6931
0.0050	0.5000	0.0005	2	0.7900	0.7022
0.0050	0.5000	0.0010	0	0.8000	0.7920
0.0050	0.5000	0.0010	1	0.8020	0.7830
0.0050	0.5000	0.0010	2	0.7960	0.7917
0.0050	0.6000	0.0005	0	0.7980	0.7012
0.0050	0.6000	0.0005	1	0.7960	0.7028
0.0050	0.6000	0.0005	2	0.8000	0.7031
0.0050	0.6000	0.0010	0	0.7980	0.7881
0.0050	0.6000	0.0010	1	0.7960	0.7929
0.0050	0.6000	0.0010	2	0.7960	0.7915
0.0100	0.5000	0.0005	0	0.8080	0.6892
0.0100	0.5000	0.0005	1	0.8000	0.6983
0.0100	0.5000	0.0005	2	0.8060	0.6848
0.0100	0.5000	0.0010	0	0.8020	0.7782
0.0100	0.5000	0.0010	1	0.8000	0.7857
0.0100	0.5000	0.0010	2	0.7980	0.7933
0.0100	0.6000	0.0005	0	0.8000	0.6889
0.0100	0.6000	0.0005	1	0.7960	0.6931
0.0100	0.6000	0.0005	2	0.8020	0.6986
0.0100	0.6000	0.0010	0	0.8000	0.7772
0.0100	0.6000	0.0010	1	0.8000	0.7806
0.0100	0.6000	0.0010	2	0.7960	0.7981

Appendix C: Depth Sweep Per-Seed Raw Accuracy

The 150 individual runs (GCN, GraphSAGE, GAT x five depths x ten seeds, unmitigated) behind the aggregate in Table 3 (Section 6.3).

Table 12: Unmitigated depth sweep, per-seed test accuracy and macro-F1.

convType	numLayers	seed	testAccuracy	testMacroF1
gat	2	0	0.7980	0.7845
gat	2	1	0.7670	0.7554
gat	2	2	0.7890	0.7748
gat	2	3	0.7830	0.7859
gat	2	4	0.8170	0.8056
gat	2	5	0.7900	0.7738
gat	2	6	0.7880	0.7746
gat	2	7	0.7950	0.7833
gat	2	8	0.7630	0.7542
gat	2	9	0.8000	0.7879
gat	4	0	0.7560	0.7522
gat	4	1	0.7800	0.7688
gat	4	2	0.7600	0.7461
gat	4	3	0.7860	0.7827
gat	4	4	0.7790	0.7737
gat	4	5	0.8030	0.7945
gat	4	6	0.7540	0.7454
gat	4	7	0.7310	0.7359
gat	4	8	0.7040	0.7241
gat	4	9	0.5970	0.5677
gat	8	0	0.2190	0.0835
gat	8	1	0.1490	0.0371
gat	8	2	0.3190	0.0691
gat	8	3	0.3050	0.2461
gat	8	4	0.3190	0.0691
gat	8	5	0.4730	0.4638
gat	8	6	0.3190	0.0691
gat	8	7	0.3190	0.0691
gat	8	8	0.3190	0.0691
gat	8	9	0.4870	0.3873
gat	16	0	0.3190	0.0691
gat	16	1	0.3190	0.0691
gat	16	2	0.3190	0.0691
gat	16	3	0.3190	0.0691
gat	16	4	0.3190	0.0691
gat	16	5	0.3190	0.0691
gat	16	6	0.1490	0.0371
gat	16	7	0.3190	0.0691
gat	16	8	0.3190	0.0691
gat	16	9	0.1440	0.0360

convType	numLayers	seed	testAccuracy	testMacroF1
gat	32	0	0.1440	0.0360
gat	32	1	0.1300	0.0329
gat	32	2	0.1490	0.0371
gat	32	3	0.1440	0.0360
gat	32	4	0.1490	0.0371
gat	32	5	0.3190	0.0691
gat	32	6	0.1440	0.0360
gat	32	7	0.1490	0.0371
gat	32	8	0.3190	0.0691
gat	32	9	0.3190	0.0691
gcn	2	0	0.8140	0.8017
gcn	2	1	0.8100	0.7980
gcn	2	2	0.8140	0.8032
gcn	2	3	0.8130	0.8035
gcn	2	4	0.8170	0.8038
gcn	2	5	0.8200	0.8081
gcn	2	6	0.8200	0.8087
gcn	2	7	0.8130	0.8015
gcn	2	8	0.8140	0.8020
gcn	2	9	0.8140	0.8013
gcn	4	0	0.7730	0.7680
gcn	4	1	0.7050	0.6972
gcn	4	2	0.7400	0.7380
gcn	4	3	0.7790	0.7720
gcn	4	4	0.5620	0.6116
gcn	4	5	0.7820	0.7711
gcn	4	6	0.7540	0.7560
gcn	4	7	0.7560	0.7553
gcn	4	8	0.7450	0.7516
gcn	4	9	0.7720	0.7597
gcn	8	0	0.4920	0.3964
gcn	8	1	0.4020	0.3254
gcn	8	2	0.4220	0.3539
gcn	8	3	0.4300	0.3913
gcn	8	4	0.2380	0.2222
gcn	8	5	0.1750	0.1201
gcn	8	6	0.2350	0.2188
gcn	8	7	0.2300	0.1932
gcn	8	8	0.2670	0.2326
gcn	8	9	0.4580	0.4098
gcn	16	0	0.2440	0.2277
gcn	16	1	0.2620	0.2390
gcn	16	2	0.2230	0.2052
gcn	16	3	0.2310	0.1954
gcn	16	4	0.2590	0.2429
gcn	16	5	0.2300	0.2110

convType	numLayers	seed	testAccuracy	testMacroF1
gcn	16	6	0.2420	0.2086
gcn	16	7	0.2710	0.2092
gcn	16	8	0.2450	0.2308
gcn	16	9	0.1450	0.1344
gcn	32	0	0.2210	0.1830
gcn	32	1	0.2450	0.2229
gcn	32	2	0.2570	0.2336
gcn	32	3	0.2380	0.1969
gcn	32	4	0.2570	0.2168
gcn	32	5	0.2120	0.1894
gcn	32	6	0.2620	0.2094
gcn	32	7	0.2440	0.2104
gcn	32	8	0.2340	0.2162
gcn	32	9	0.2370	0.2237
sage	2	0	0.8020	0.7920
sage	2	1	0.8050	0.7959
sage	2	2	0.8030	0.7923
sage	2	3	0.7980	0.7894
sage	2	4	0.8000	0.7892
sage	2	5	0.7910	0.7836
sage	2	6	0.8030	0.7954
sage	2	7	0.8020	0.7910
sage	2	8	0.8000	0.7905
sage	2	9	0.8060	0.7974
sage	4	0	0.6580	0.6627
sage	4	1	0.7410	0.7296
sage	4	2	0.7210	0.6882
sage	4	3	0.7370	0.7335
sage	4	4	0.6580	0.6733
sage	4	5	0.7030	0.7043
sage	4	6	0.7120	0.7293
sage	4	7	0.6780	0.6266
sage	4	8	0.8060	0.7948
sage	4	9	0.7580	0.7436
sage	8	0	0.3190	0.0691
sage	8	1	0.3190	0.0691
sage	8	2	0.3190	0.0691
sage	8	3	0.0910	0.0238
sage	8	4	0.1420	0.1501
sage	8	5	0.3190	0.0691
sage	8	6	0.1440	0.0360
sage	8	7	0.3190	0.0691
sage	8	8	0.4310	0.2771
sage	8	9	0.3190	0.0691
sage	16	0	0.3190	0.0691
sage	16	1	0.3190	0.0691

convType	numLayers	seed	testAccuracy	testMacroF1
sage	16	2	0.3190	0.0691
sage	16	3	0.1490	0.0371
sage	16	4	0.3190	0.0691
sage	16	5	0.3190	0.0691
sage	16	6	0.3190	0.0691
sage	16	7	0.3190	0.0691
sage	16	8	0.3190	0.0691
sage	16	9	0.1490	0.0371
sage	32	0	0.1490	0.0371
sage	32	1	0.3190	0.0691
sage	32	2	0.1030	0.0267
sage	32	3	0.1300	0.0329
sage	32	4	0.3190	0.0691
sage	32	5	0.3190	0.0691
sage	32	6	0.3190	0.0691
sage	32	7	0.1440	0.0360
sage	32	8	0.3190	0.0691
sage	32	9	0.1440	0.0360

Appendix D: Depth-32 Unmitigated GCN Per-Layer Checkpoint Energy

The complete per-seed, per-layer per-dimension Dirichlet energy behind the below-floor counts in Table 7 (Section 6.5), for all 31 comparable-band layers across all 10 seeds. `belowFloor` marks whether the value is below the `1e-12` floor Section 4.5 describes; energy is reported in scientific notation, since it spans roughly 30 orders of magnitude across this table and a fixed decimal count would round every collapsed layer to zero.

Table 13: Checkpoint per-dimension Dirichlet energy, unmitigated GCN, depth 32, every band layer, all 10 seeds.

seed	layer	dirichletEnergy	belowFloor
0	1	1.44e-15	True
0	2	8.66e-16	True
0	3	7.24e-16	True
0	4	1.25e-15	True
0	5	1.69e-15	True
0	6	1.18e-15	True
0	7	2.47e-15	True
0	8	2.42e-15	True
0	9	4.61e-15	True
0	10	3.01e-15	True
0	11	4.79e-15	True
0	12	5.84e-15	True
0	13	9.45e-15	True
0	14	4.73e-15	True
0	15	1.61e-14	True
0	16	1.86e-14	True
0	17	2.86e-14	True
0	18	2.37e-14	True
0	19	2.10e-14	True
0	20	1.10e-13	True
0	21	3.67e-14	True
0	22	9.76e-14	True
0	23	3.83e-12	False
0	24	2.24e-11	False
0	25	2.24e-07	False
0	26	9.56e-03	False
0	27	3.50e-03	False
0	28	5.55e-01	False
0	29	1.09e+00	False
0	30	1.52e+00	False
0	31	7.16e+00	False
1	1	5.09e-20	True
1	2	4.72e-20	True
1	3	6.76e-20	True

seed	layer	dirichletEnergy	belowFloor
1	4	1.09e-19	True
1	5	1.54e-19	True
1	6	3.22e-19	True
1	7	4.47e-19	True
1	8	1.87e-19	True
1	9	1.00e-18	True
1	10	4.12e-19	True
1	11	6.41e-18	True
1	12	7.82e-19	True
1	13	3.55e-19	True
1	14	1.60e-18	True
1	15	7.55e-19	True
1	16	6.07e-19	True
1	17	1.12e-18	True
1	18	2.86e-18	True
1	19	1.04e-18	True
1	20	2.01e-14	True
1	21	1.75e-14	True
1	22	1.35e-15	True
1	23	2.11e-12	False
1	24	1.82e-08	False
1	25	5.72e-06	False
1	26	7.66e-10	False
1	27	4.54e-09	False
1	28	9.88e-01	False
1	29	1.32e-01	False
1	30	1.44e+00	False
1	31	2.04e+01	False
2	1	7.49e-33	True
2	2	1.31e-32	True
2	3	8.36e-33	True
2	4	9.02e-33	True
2	5	1.18e-32	True
2	6	1.39e-32	True
2	7	1.38e-32	True
2	8	1.06e-32	True
2	9	1.95e-32	True
2	10	3.01e-32	True
2	11	4.47e-32	True
2	12	2.15e-32	True
2	13	3.54e-32	True
2	14	4.96e-32	True
2	15	6.13e-32	True
2	16	6.89e-32	True
2	17	1.01e-31	True
2	18	9.64e-32	True

seed	layer	dirichletEnergy	belowFloor
2	19	1.16e-31	True
2	20	1.94e-31	True
2	21	7.08e-31	True
2	22	1.60e-30	True
2	23	9.10e-31	True
2	24	2.95e-28	True
2	25	1.06e-20	True
2	26	0.00e+00	True
2	27	7.41e-04	False
2	28	9.98e-01	False
2	29	1.45e+00	False
2	30	2.62e+00	False
2	31	1.25e+01	False
3	1	6.50e-24	True
3	2	2.91e-24	True
3	3	3.65e-24	True
3	4	5.57e-24	True
3	5	9.20e-24	True
3	6	1.02e-23	True
3	7	1.76e-23	True
3	8	1.80e-23	True
3	9	1.59e-23	True
3	10	9.70e-24	True
3	11	2.66e-23	True
3	12	4.58e-23	True
3	13	3.93e-23	True
3	14	5.38e-23	True
3	15	6.92e-23	True
3	16	7.02e-23	True
3	17	6.60e-23	True
3	18	9.21e-23	True
3	19	6.66e-23	True
3	20	1.28e-22	True
3	21	2.93e-22	True
3	22	6.33e-22	True
3	23	9.33e-22	True
3	24	7.90e-13	True
3	25	2.13e-11	False
3	26	3.17e-05	False
3	27	1.51e-04	False
3	28	7.25e-01	False
3	29	7.26e-01	False
3	30	1.38e+00	False
3	31	1.14e+01	False
4	1	2.86e-35	True
4	2	2.30e-35	True

seed	layer	dirichletEnergy	belowFloor
4	3	2.07e-35	True
4	4	1.98e-35	True
4	5	2.58e-35	True
4	6	2.67e-35	True
4	7	3.15e-35	True
4	8	4.30e-35	True
4	9	5.84e-35	True
4	10	9.43e-35	True
4	11	1.03e-34	True
4	12	9.38e-35	True
4	13	6.81e-35	True
4	14	3.03e-34	True
4	15	2.16e-34	True
4	16	6.09e-34	True
4	17	3.77e-34	True
4	18	1.28e-33	True
4	19	6.00e-32	True
4	20	1.70e-33	True
4	21	6.34e-33	True
4	22	2.15e-32	True
4	23	1.01e-30	True
4	24	8.66e-28	True
4	25	3.16e-12	False
4	26	4.78e-04	False
4	27	0.00e+00	True
4	28	1.15e+00	False
4	29	1.21e+00	False
4	30	3.14e+00	False
4	31	2.45e+01	False
5	1	1.89e-31	True
5	2	2.02e-31	True
5	3	2.64e-31	True
5	4	2.36e-31	True
5	5	4.41e-31	True
5	6	1.92e-31	True
5	7	4.41e-31	True
5	8	3.14e-31	True
5	9	4.82e-31	True
5	10	3.82e-31	True
5	11	1.07e-30	True
5	12	7.83e-31	True
5	13	1.23e-30	True
5	14	7.08e-31	True
5	15	1.13e-30	True
5	16	2.31e-30	True
5	17	2.67e-30	True

seed	layer	dirichletEnergy	belowFloor
5	18	2.74e-30	True
5	19	3.62e-30	True
5	20	5.68e-30	True
5	21	4.53e-30	True
5	22	1.54e-28	True
5	23	1.41e-26	True
5	24	1.88e-26	True
5	25	4.27e-20	True
5	26	2.42e-11	False
5	27	1.43e-05	False
5	28	1.81e-02	False
5	29	1.97e+00	False
5	30	2.87e+00	False
5	31	7.02e+00	False
6	1	3.28e-12	False
6	2	2.09e-12	False
6	3	2.40e-12	False
6	4	2.14e-12	False
6	5	4.18e-12	False
6	6	5.31e-12	False
6	7	4.50e-12	False
6	8	7.72e-12	False
6	9	8.92e-12	False
6	10	1.79e-11	False
6	11	9.53e-12	False
6	12	1.62e-11	False
6	13	1.44e-11	False
6	14	1.25e-11	False
6	15	2.35e-11	False
6	16	2.86e-11	False
6	17	4.60e-11	False
6	18	4.36e-11	False
6	19	8.77e-10	False
6	20	1.19e-10	False
6	21	7.05e-10	False
6	22	8.93e-10	False
6	23	1.93e-09	False
6	24	2.34e-08	False
6	25	7.14e-05	False
6	26	1.45e-02	False
6	27	2.39e-07	False
6	28	4.78e-01	False
6	29	4.57e-01	False
6	30	9.76e-01	False
6	31	1.06e+01	False
7	1	1.50e-26	True

seed	layer	dirichletEnergy	belowFloor
7	2	1.73e-26	True
7	3	1.05e-26	True
7	4	2.81e-26	True
7	5	2.05e-26	True
7	6	3.27e-26	True
7	7	2.13e-26	True
7	8	4.21e-26	True
7	9	5.09e-26	True
7	10	1.20e-25	True
7	11	7.72e-26	True
7	12	8.58e-26	True
7	13	1.20e-25	True
7	14	9.65e-26	True
7	15	2.45e-25	True
7	16	1.97e-25	True
7	17	1.39e-25	True
7	18	1.50e-25	True
7	19	2.65e-25	True
7	20	6.61e-24	True
7	21	5.44e-24	True
7	22	2.04e-23	True
7	23	1.14e-21	True
7	24	6.07e-10	False
7	25	1.83e-15	True
7	26	4.51e-07	False
7	27	0.00e+00	True
7	28	6.52e-01	False
7	29	1.28e-01	False
7	30	1.02e+00	False
7	31	1.38e+01	False
8	1	4.23e-28	True
8	2	8.62e-29	True
8	3	3.39e-28	True
8	4	5.78e-28	True
8	5	5.87e-28	True
8	6	3.63e-28	True
8	7	1.59e-27	True
8	8	7.82e-28	True
8	9	1.07e-27	True
8	10	1.31e-27	True
8	11	1.89e-27	True
8	12	2.07e-27	True
8	13	2.39e-27	True
8	14	3.09e-27	True
8	15	3.83e-27	True
8	16	5.04e-27	True

seed	layer	dirichletEnergy	belowFloor
8	17	3.72e-27	True
8	18	7.17e-27	True
8	19	1.12e-26	True
8	20	2.30e-26	True
8	21	1.08e-26	True
8	22	3.74e-08	False
8	23	4.75e-23	True
8	24	1.69e-20	True
8	25	9.22e-20	True
8	26	2.34e-12	False
8	27	1.28e-09	False
8	28	0.00e+00	True
8	29	1.58e+00	False
8	30	2.20e+00	False
8	31	7.14e+00	False
9	1	2.87e-38	True
9	2	2.61e-38	True
9	3	2.14e-38	True
9	4	5.37e-38	True
9	5	7.30e-38	True
9	6	2.16e-38	True
9	7	5.55e-38	True
9	8	1.24e-37	True
9	9	1.06e-37	True
9	10	9.06e-38	True
9	11	8.84e-38	True
9	12	2.59e-37	True
9	13	3.71e-37	True
9	14	4.18e-37	True
9	15	5.41e-37	True
9	16	6.02e-37	True
9	17	6.12e-37	True
9	18	1.10e-36	True
9	19	1.28e-36	True
9	20	6.00e-36	True
9	21	6.78e-36	True
9	22	5.31e-35	True
9	23	1.04e-34	True
9	24	6.20e-33	True
9	25	2.12e-29	True
9	26	1.03e-16	True
9	27	1.50e-21	True
9	28	2.07e-02	False
9	29	1.81e+00	False
9	30	2.82e+00	False

9	31	7.16e+00	False